Electronic Will Application
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By

[illegible]	[illegible]
[illegible]	
[illegible]	
[illegible]	
Rahaf Hassan Ali	
[illegible]	

Supervised by:

Dr. Sara Abdulrahman

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COMMITTEE REPORT

We certify that we have read this graduation project report as examining committee examined the student in its content and that in our opinion it is adequate as a project document for B.Sc. in Computer Science

Supervisor**:**	Examiner 1**:**	Examiner 2**:**
Dr.	Name:	Name:
Signature:	Signature:	Signature:
Date:/ /	Date:/ /	Date:/ /

ABSTRACT

The Electronic Will Application is a comprehensive application designed to assist users in creating and managing their wills and testamentary instructions efficiently and securely. This innovative system provides a user-friendly platform for individuals to document their financial assets, properties, and posthumous wishes, ensuring a seamless transition of their estate to beneficiaries upon their passing.

Users can create accounts, enter detailed information about their assets, and specify the intended recipients of their assets and possessions. The system incorporates advanced security measures, including encryption and access controls, to safeguard sensitive data and guarantee that only authorized individuals can access the wills. One of the distinctive features of this system is its integration with government databases, allowing for real-time monitoring of user status. In the event of a user's death, the system automatically sends notifications to designated individuals, providing them with a unique access code to retrieve and execute the will.

The Electronic Will Application addresses legal and ethical considerations by adhering to local laws and regulations related to wills and data privacy. It is designed to be scalable, accommodating a growing user base, and includes robust data backup and recovery mechanisms to prevent data loss. Management System aims to simplify the often complex and emotional process of estate planning, ensuring that individuals' final wishes are carried out with precision and care.

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Chapter 1 Introduction

1.1 Introduction

In an era marked by the influence of technology, traditional paradigms of life and legacy planning are undergoing a transformative evolution. The Electronic Will Application represents a pioneering venture into this new landscape, where the age-old practice of drafting and executing wills converges with cutting-edge digital solutions to create a comprehensive, secure, and user-centric platform.

The importance of preparing a will cannot be overstated. For centuries, individuals have sought to delineate the future of their assets, properties, and cherished belongings, ensuring that their legacies endure as intended. Yet, despite the significance of this endeavor, the process has often been fraught with complexity, logistical challenges, and security concerns. Electronic Will Application is an innovative application designed to redefine and streamline the entire will creation and execution experience.

This system offers users an intuitive and accessible means to draft, store, and manage their wills within a secure digital environment. The platform is engineered to provide clarity, simplicity, and peace of mind to individuals at every stage of life, regardless of their familiarity with legal nuances or technological proficiency.

One of the system's distinctive features is its seamless integration with government databases, a crucial innovation that enhances the reliability and timeliness of will execution. In a world where personal information intersects with official records, the system automatically cross-references user data with vital statistics, ensuring that wills are activated promptly upon the user's passing. This integration is not only a testament to the platform's technological sophistication but also a profound commitment to efficiency and reliability. Security and privacy stand as paramount pillars of the Electronic Will Application. In an era marked by increasing concerns about data breaches and identity theft, the application employs state-of-the-art encryption protocols and access controls to safeguard user information and testamentary documents. The assurance that sensitive details remain confidential and that wills are executed precisely as envisioned offers users a level of trust and confidence that is unparalleled in traditional will management practices.

This project aspires not merely to modernize the process of will creation and execution but to set a new standard for testamentary planning in the digital age.

1.2 Problem statement

In today's digital world, making and managing wills the old-fashioned way has become outdated and problematic. This creates several issues, such as making it hard for people to understand the legal jargon, find their wills when needed, keep them safe, and ensure that their wishes are followed after they pass away.

Additionally, there are security concerns, and it's challenging to keep up with different legal rules in various places. All these problems can lead to disputes, delays, and misunderstandings when it comes to executing wills. We need a modern solution that simplifies will management, making it easy for anyone to create, store, and execute their wills securely. This system should also work seamlessly with government records and be user-friendly for everyone, regardless of their tech skills. Such a solution will empower individuals to have better control over their legacies in the digital age.

1.3 Proposed solution

The proposed solution is an Electronic Will Application, which is an innovative software application designed to streamline and modernize the process of creating, storing, and executing wills. This system aims to address the challenges and inefficiencies associated with traditional will management by offering the following key features:

- Simplified Will Creation: The system provides an intuitive and user-friendly interface that simplifies the process of creating a will. It eliminates legal problems and complexities, making it accessible to individuals without legal expertise.
- Leaving a good legacy (Athar): The user can add to his will request to be fulfilled such as making some charity works for him, and may include the account number for this.

- Help Button: The help button provides a detailed explanation of the Islamic method for writing a will. This button assists users with limited knowledge in the Islamic way of composing a will.
- Donating medical stuff: The system has a section for donating the medical equipment that belongs to the person. The family can add these things to be donated such as wheelchair, oxygen tank, and other medical equipment.
- Efficient Will Storage and Retrieval: Users can securely store their wills digitally within the system. This eliminates the need for physical documents and makes it easy for beneficiaries and executors to access wills promptly when required.
- Enhanced Security and Privacy: The application employs robust encryption and access control measures to protect users' sensitive testamentary information, ensuring that wills remain confidential and secure.
- Preservation of Testamentary Intent: The system emphasizes clear documentation of users' final wishes, reducing the risk of disputes and ensuring that testators' intentions are accurately upheld.
- Integration with Government Records: The system seamlessly connects with government databases and vital records. It monitors the status of users and automatically notifies authorized individuals when the testator passes away, streamlining the execution process.
- User-Friendly Interface: The application is designed with a focus on user-friendliness, ensuring that individuals of varying technological backgrounds can easily create, update, and manage their wills digitally.

1.4 Objectives

The main objectives for this project are:

- Make the process of creating a will more straightforward and accessible to individuals of all backgrounds, eliminating legal complexities and jargon.
- Provide a secure digital platform for storing wills, ensuring easy access for beneficiaries and executors when needed, thus eliminating the need for physical documents.

- Implement robust security measures, including encryption and access controls, to safeguard users' sensitive testamentary information and maintain the confidentiality of wills.
- Connect with government databases and vital records to monitor the status of users and automatically notify authorized individuals upon the testator's passing, streamlining the execution process.
- Design the application with a user-centric approach, ensuring that individuals with varying levels of technological proficiency can easily create, update, and manage their wills digitally.
- Expedite the process of executing wills by automating notifications and verification procedures, reducing delays and potential errors.
- Implement advanced data security measures to protect user information and testamentary documents from unauthorized access, data breaches, and tampering.
- Provide a one-stop platform for all aspects of will management, from initial creation and storage to efficient execution, reducing the need for multiple services and intermediaries.

1.5 Scope

The Electronic Will Application is designed to make creating and managing wills easy for users in Saudi Arabia and Arabic-speaking regions, but it can be used anywhere. It supports both Arabic and English languages and offers features like user registration, will creation, legal guidance, secure storage, and more. It's user-friendly and secure, ensuring that users can efficiently manage their wills, regardless of their location or language preference.

1.6 Hardware and Software Requirement

1.6.1 Software Requirements:

The Electronic Will Application relies on a variety of software tools and platforms for development and operation. These include:

- Android Studio IDE: This integrated development environment is used for building the application, utilizing Java programming language and XML for Android app development.
- MySQL: The database management system (DBMS) responsible for storing and managing user data and will information.
- Php Apache Server: This server is essential for running PHP pages, which are used for connecting the application to the MySQL database.
- XAMPP Server: XAMPP is utilized to initiate and manage the MySQL DBMS and PHP Apache server, creating a local development environment.
- Notepad++: An editor used for writing PHP code, facilitating the development of server-side components.

1.6.2 Hardware Requirements:

The Electronic Will Application also has specific hardware requirements:

- Android Mobile Device: To run and test the application, an Android mobile device is required, ensuring that the developed app functions as intended on real devices.
- Internet Access: An internet connection is necessary for accessing online resources, testing remote features, and interacting with the database server.
- Personal Computer (Server): A personal computer is used as a server for testing purposes, with the following minimum specifications:
    - Processor: Intel Core i5 or equivalent.
    - Hard Disk: Minimum 1 TB storage capacity.
    - RAM: At least 8 GB RAM for efficient operation.

These hardware and software requirements collectively support the development, testing, and operation of the Electronic Will Application, ensuring its functionality and usability.

Chapter 2 Literature Survey

2.1 Overview

In an age marked by the increasing digitalization of our lives, the realm of estate planning and will management has undergone a profound transformation. The traditional methods of drafting wills on paper and storing them in secure vaults have given way to digital estate planning, reflecting the evolving nature of assets and the need for efficient, secure, and accessible solutions. This literature review serves as an exploration into the multifaceted landscape of digital estate planning and the development of the Electronic Will Application. The topic of digital wills is relatively new, and there is not a lot of research on the subject. However, there are a few studies that have been conducted, including some in the Arabian countries.

One study [1] found that 58% of adults in the UAE would be interested in creating a digital will. The study also found that the main reasons for creating a digital will were to ensure that the deceased's wishes are respected and to make it easier for their loved ones to deal with their estate. Another study [2] found that digital wills can be a valuable tool for people who have complex estates or who have assets located in multiple jurisdictions. The study also found that digital wills can be a cost-effective and convenient way to create a will.

A third study [3] found that the use of digital wills is increasing in Kuwait. The study also found that the main reason for the increase is the growing awareness of the benefits of digital wills. A fourth study [4] reviewed the literature on digital wills to identify their potential advantages and disadvantages. study found that digital wills offer a number of advantages over traditional wills, including security, convenience, and accessibility. However, it also noted that there are some challenges associated with digital wills, such as legal compliance and security concerns. A fifth study [5] predicted the future of digital wills based on current trends and technological advancements. study predicted that digital

wills will become increasingly popular in the future, due to the technological advancements and the growing awareness of their benefits.

While there is limited research on the topic of digital wills, the existing research suggests that digital wills can be a valuable tool for people who want to make sure that their wishes are respected after they die. These studies collectively underscore the growing importance of digital estate planning and the pivotal role of technology in addressing its challenges. They emphasize the need for user-centric design, data security, cross-border compatibility, and government integration. Additionally, studies from the Arab world highlight the cultural and regional nuances that must be considered in developing effective digital will management solutions. These findings inform the development of the Electronic Will Application, aiming to offer a secure, user-friendly, and culturally sensitive platform for individuals seeking to manage their digital legacies.

Despite the challenges, the studies cited suggest that digital wills are likely to become more popular in the Arabian countries in the future. This is due to a number of factors, including the increasing use of technology in the region and the growing awareness of the benefits of digital wills. The studies suggest that digital wills are becoming increasingly popular around the world, including in the Arab world, and the legal changes in some countries that have made digital wills more widely accepted.

2.2 Similar Applications

2.2.1 Deliver My Will App "وصل وصيتي"

This app is an Arabic-language mobile application that allows users to create and manage their wills. The app is available on the Google Play Store. Here is a detailed description of the app's features:

- Will creation: The app allows users to create a will from scratch or by using a template. Users can specify their assets, liabilities, and beneficiaries, as well as their wishes for their funeral and burial.

- Will management: Users can edit, update, and delete their wills at any time. They can also share their wills with their loved ones or with a lawyer.
- Islamic law compliance: The app allows users to create a will in accordance with Islamic law. This is an important feature for many Muslims, as it ensures that their assets will be distributed according to their religious beliefs.

Pros:

- The app is easy to use and navigate.
- The app is available in Arabic, which makes it accessible to a wide range of users.
- The app allows users to create a will in accordance with Islamic law.
- The app includes a number of additional features that are useful for estate planning.

Cons:

- The app is currently only available on Android devices.
- The app is not as well-known applications for managing wills, and not used by users.
- The app is not connected with the government records directly.
- It doesn’t send notifications to authorized users.

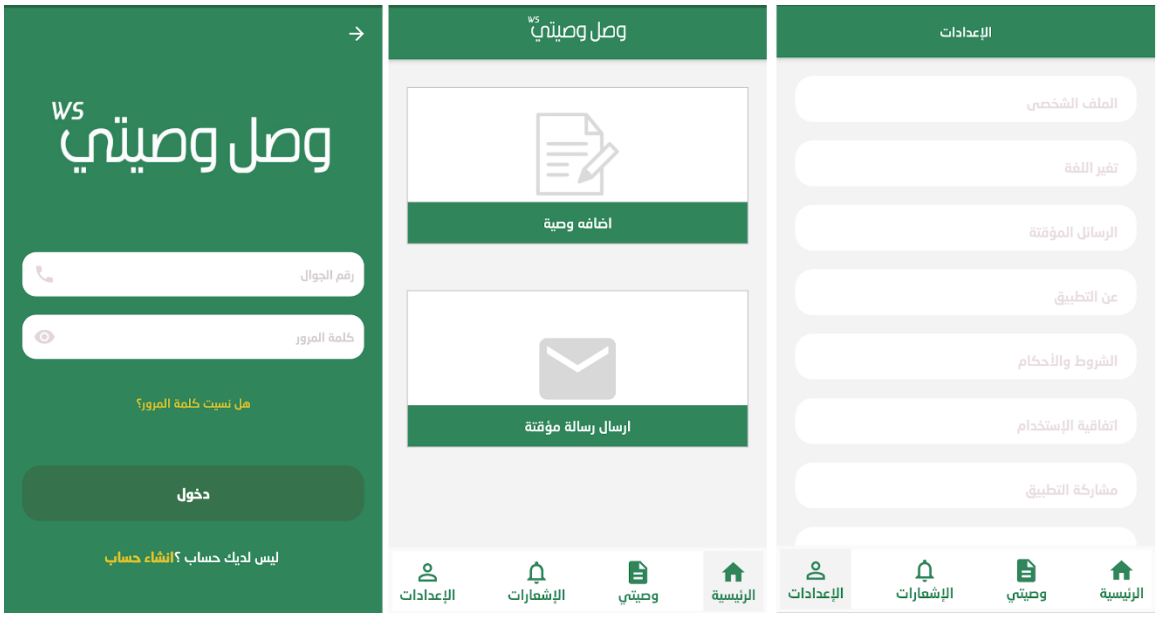

Figure 2.1 Deliver My Will App "وصل وصيتي".

2.2.2 The Will App "الوصية"

This app is a mobile application which is available on the Apple App Store. It is an Arabic-language app that allows users to create and manage their wills. The app also includes a number of features that are specific to the Arab world, such as the ability to create a will in accordance with Islamic law. The application enables you to:

- Create a simplified will and request consultation or share it with others.
- Create a detailed will, including:
- Listing assets (bank accounts and real estate).
- Documenting debts and rights (rights of individuals and the right of Allah Almighty).
- Registering deposits and trusts (company shares, cars, real estate contributions).
- Appoint a supervisor for the endowment (waqf).
- Appoint an executor for the will.
- Recommended actions (recommended spending, marriages, minor children).

Pros:

- The app is easy to use and navigate.
- The app is available in Arabic, which makes it accessible to a wide range of users.
- The app allows users to document debts and rights, register deposits and trusts and create a will in accordance with Islamic law.
- Supported by AL BILAD BANK

Cons:

- The app is currently only available on iOS devices.
- The app is not well known or popular.
- The app requires the users to upload the death certificates manually.
- It’s not connected to the government records.

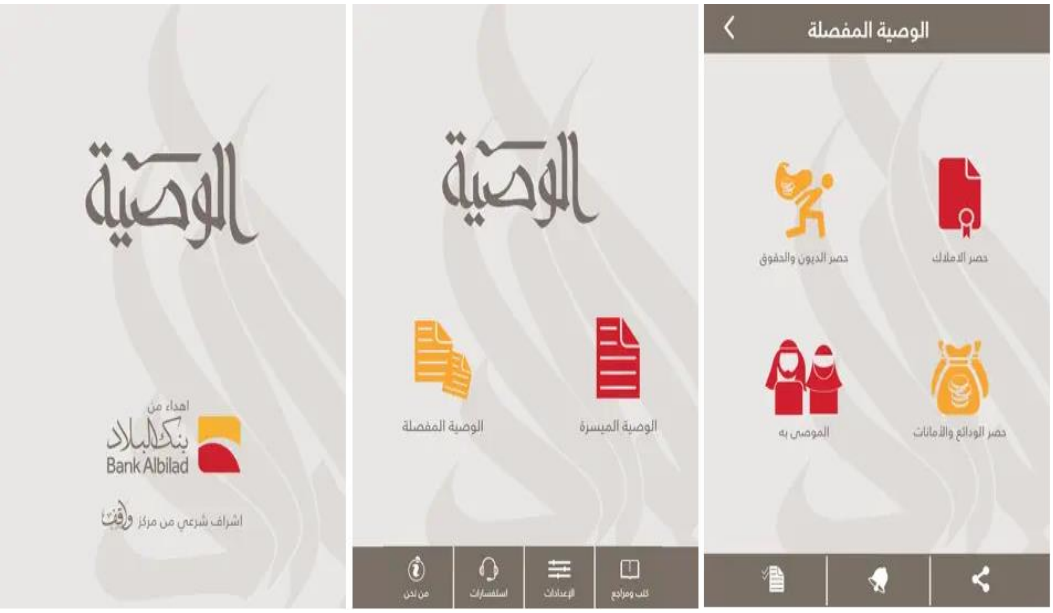

Figure 2.2 The Will App "الوصية"

2.2.3 WillMaker

WillMaker is a mobile app that allows users to create, edit, and share their wills from the convenience of their smartphones. The app is available in both English and Spanish. WillMaker's interface is simple and easy to use. Users can start by creating a new will or

by using a template. The app then guides users through a series of questions to collect information about their assets, liabilities, beneficiaries, and executor. Once the will is complete, users can review it and make any necessary changes. They can then share the will with their loved ones or with a lawyer. It also includes a number of features that can be helpful for people with complex estates. For example, users can create multiple wills, set up trusts, and designate guardians for their minor children.

Pros:

- Create a will from scratch or use a template
- Specify your assets, liabilities, and beneficiaries
- Choose your executor and guardians
- Share your will with your loved ones or with a lawyer
- Easy to use
- Available in two languages

Cons:

- It doesn’t support the Arabic language.
- It doesn’t include creating a will in accordance with Islamic law.
- It’s not connected to the government records.
- It doesn’t send notifications to the authorized people after the death of the will owner.
- Not free.

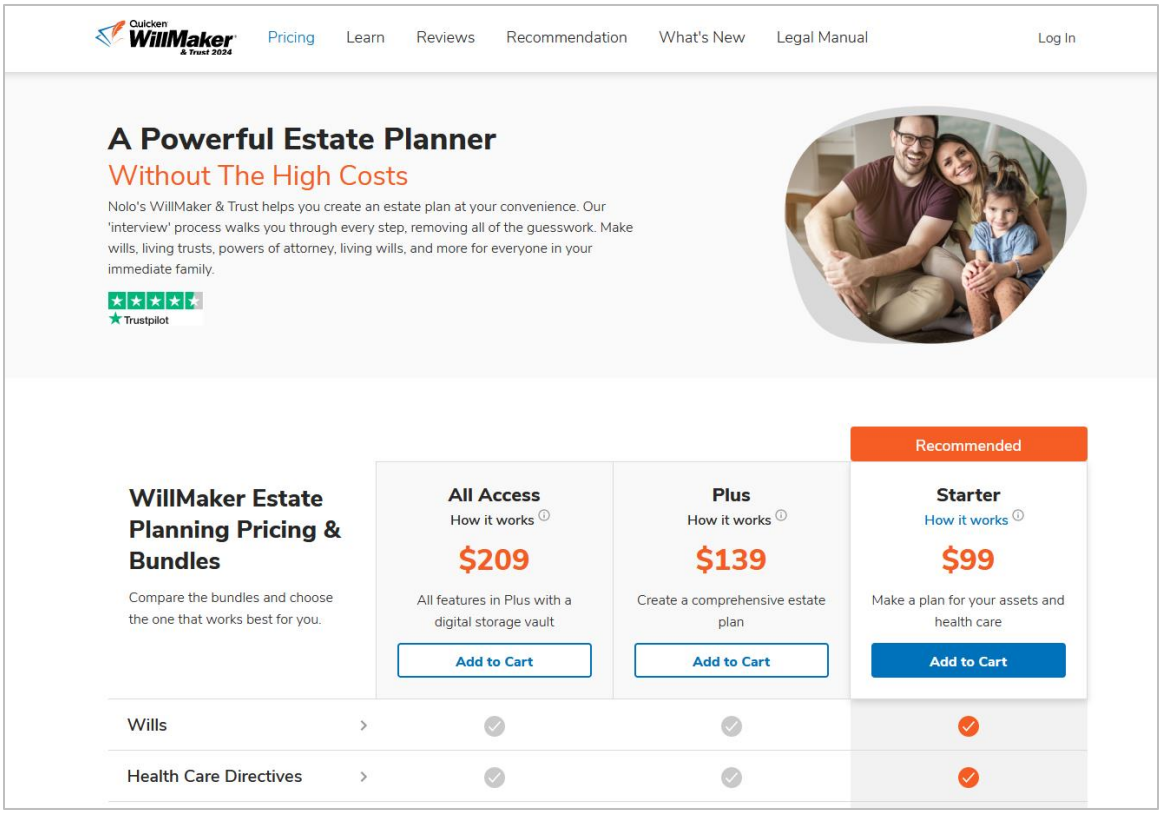

Figure 2.3 Willmaker interface.

2.2.4 Trust & Will

Trust & Will is a mobile app and website that allows users to create and manage their wills, trusts, and other estate planning documents. The app is available in English and Spanish. In addition to creating wills, users can create trusts, living wills, and other estate planning documents.

Pros:

- Create a will from scratch or use a template
- Specify your assets, liabilities, and beneficiaries
- Choose your executor and guardians
- Create a trust for your minor children or other loved ones

- Store important documents, such as your death certificate and a copy of your will

Con:

- It doesn't support the Arabic language.
- Not easy to use.
- Not free.
- It doesn't include creating a will in accordance with Islamic law.
- It's not connected to the government records.
- It doesn't send notifications to the authorized people after the death of the will owner.

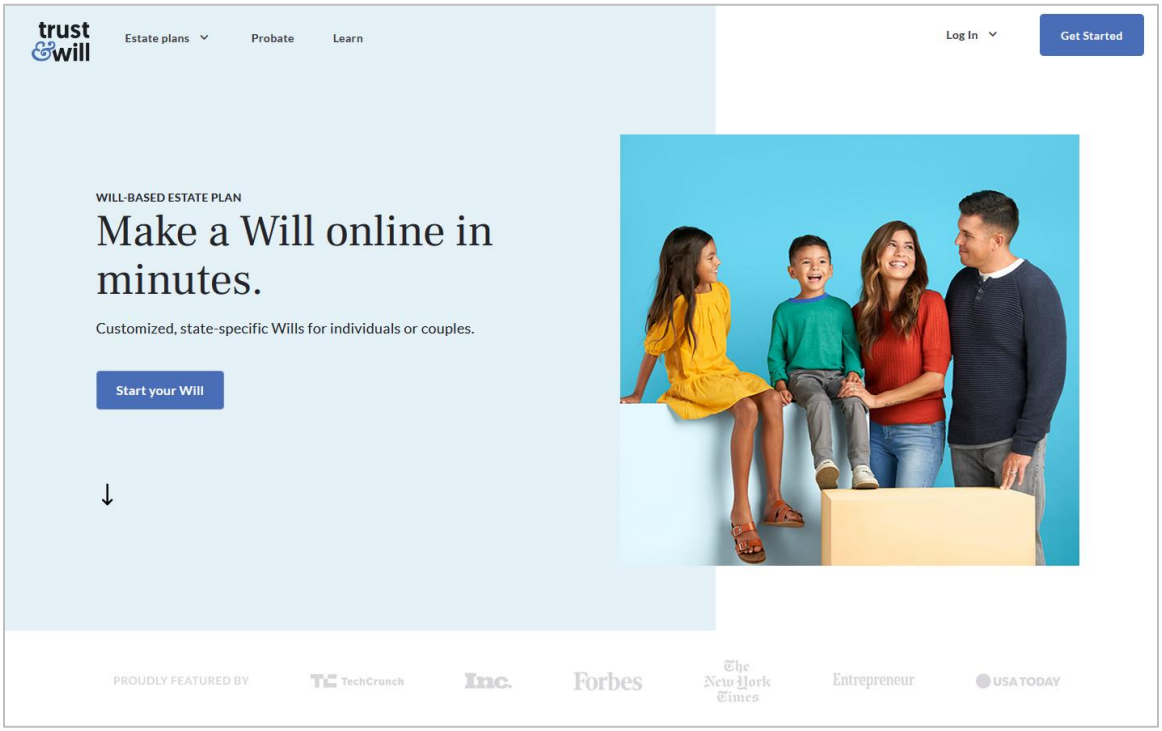

Figure 2.4 Trust & Will interface.

2.3 Conclusion

WillMaker and Trust & Will are both good options for creating a will. However, there are some key differences between the two apps. WillMaker is easier to use and more affordable than Trust & Will. However, Trust & Will offers a wider range of features, including the ability to create trusts and other estate planning documents. Neither WillMaker nor Trust & Will are currently available in Arabic. This is a significant drawback, as Arabic is the primary language of many Arab countries especially in the KSA. In addition to the specialty of the Islamic laws in the will making process. The reviewed Arabic applications are good to be used but they are missing some of the important features such as the integration with the government records to get the new deaths directly. They require the attaching of the death certificates. In addition to not sending direct notifications to the users who are authorized to pen the wills.

For conclusion we are required to build an application that provides the following services:

- Allow the users to manage their wills easily including the details about their assets, and liabilities.
- Make the application suitable for the KSA users by supporting the Arabic language interface, and making the will creation according to the Islamic law.
- Integrate the application with the government record for death cases.
- Send direct notifications to the authorized users using phone messages including a code that allows them to open the will after the death of the user.
- Keep the privacy of the users using data encryption and access control.
- Make the process simple and easy to use with a friendly interface.
- Make the application free to the users.

Chapter 3 System Analysis

3.1 Methodology

The selected methodology for the development process if the waterfall SDLC model, The Waterfall model is a traditional project management and software development approach that is characterized by a linear and sequential workflow. Here are the key phases and steps involved in the Waterfall model:

- Requirements Analysis:
  - Define and document the project requirements in detail.
  - Gather input from stakeholders to ensure a comprehensive understanding of the project scope.
- System Design:
  - Develop a detailed system design based on the requirements.
  - Design the UML diagrams and user interfaces.
  - Create design documents.
- Implementation (Coding):
  - Write code based on the design specifications.
  - Develop the software or system in accordance with the design documents.
  - Conduct unit testing to identify and fix any coding errors.
- Testing:
  - Perform system testing to verify that the software meets the specified requirements.
  - Identify and address any defects, issues, or inconsistencies.
  - Conduct user acceptance testing to ensure that the system meets the users' needs.
- Deployment:
  - Deploy the software in a production environment.
  - Make it available to end-users.
- Maintenance:

    - Provide ongoing maintenance and support to address issues, update features, and ensure the system's continued functionality.

The Waterfall model is a good choice for the project because we have well-defined, stable requirements, limited budget, and strict deadlines. Also, we are working on a small to medium-sized project, and we prefer a low-risk, sequential development approach.

3.2 Requirement Elicitation Techniques

Effective requirement elicitation techniques are essential to understand and capture the needs of users and the functionality of the application. Here are the requirement elicitation techniques for our project:

- Interviews:
    - Conduct structured interviews with potential users, legal experts, and individuals involved in will planning and the related Islamic laws.
    - Ask questions about their expectations, challenges, and what features they would like to see in the application.
    - Probe for details on how they currently manage wills and estate planning.
- User Story Mapping:
    - Create user story maps that visually represent the user's journey and interaction with the application.
    - Identify key user stories and prioritize them based on importance.
- Context Diagrams:
    - Develop context diagrams to illustrate how the application will interact with external entities, such as government databases or notification systems.
    - Define the scope and boundaries of the system.
- Document Analysis:
    - Review existing wills, legal documents, and estate planning guidelines to ensure that the application aligns with legal standards.
- Expert Consultation:
    - Consult with experts to ensure the application complies with relevant laws and regulations for will management.

3.3 Requirement Analysis

3.3.1 Functional requirements

The users of the systems are:

- **Testators (Users Creating Wills):** These are the individuals who use the application to create, update, and manage their own wills.
- **Executors or Representatives (Responsible for Applying the Will):** These users are responsible for ensuring that the terms of the will are carried out according to the testator's wishes.
- **Donors:** users who join the application to donate items and medical equipment such as wheelchair, oxygen tank, and other medical equipment.
- **Customer Support:** Individuals responsible for providing customer support and assistance to users.
- **Unregistered users:** Individuals who browse the donations in the application and request them from donors.

The functional requirements of the **Testators (Users Creating Wills):**

- Register a user account securely, including providing personal information.
- Manage user profiles, including updating contact information and personal details.
- Create a new will by providing detailed information about assets, beneficiaries, and specific details.
- List assets, including financial accounts, real estate, personal property, and digital assets. Specify the location and ownership details of assets.
- Appoint executors or representatives responsible for carrying out the terms of the will.
- Request support from technical support team if needed.

The functional requirements of the **Executors (Responsible for Applying the Will):**

- Receive notification as an EMAIL message from the system when the user passes away, the notification contains a secret code.
- Use the received code to login to their executor account.

- Access the will documents and related information for the testator they are representing. And work on applying it.

The functional requirements of **Donors:**

- Register for a user account and create and manage user profiles, including contact information and preferences.
- list items they want to donate, such as wheelchairs, oxygen tanks, and medical equipment.
- Include fields for item descriptions, quantities, conditions, and any additional notes.
- upload photos of the items.
- specify the location or pickup point for the donated items.
- schedule or request pickups or drop-offs for their donated items.
- communicate with recipient.

The functional requirements of **Customer Support:**

- Login to the system.
- View the sent support requests
- Respond to the support requests

The functional requirements of **Unregistered user:**

- view donated items.
- contact donors to get the donated items.
- Pick up the donated items from the donors.

3.3.1 Non-Functional Requirements

- **Reliability**: The system should be highly reliable, minimizing downtime and ensuring that users can access it when needed.
- **Data Encryption**: Ensure that sensitive user data and will information are encrypted to protect against unauthorized access.
- **Authentication**: Implement secure user authentication and authorization mechanisms to prevent unauthorized use of the system.

- **Availability**: The system should be available 24/7, with minimal scheduled downtime for maintenance.
- **Usability**: The user interface should be intuitive, user-friendly, and accessible for users of different skill levels.

3.4 Use Case Diagram

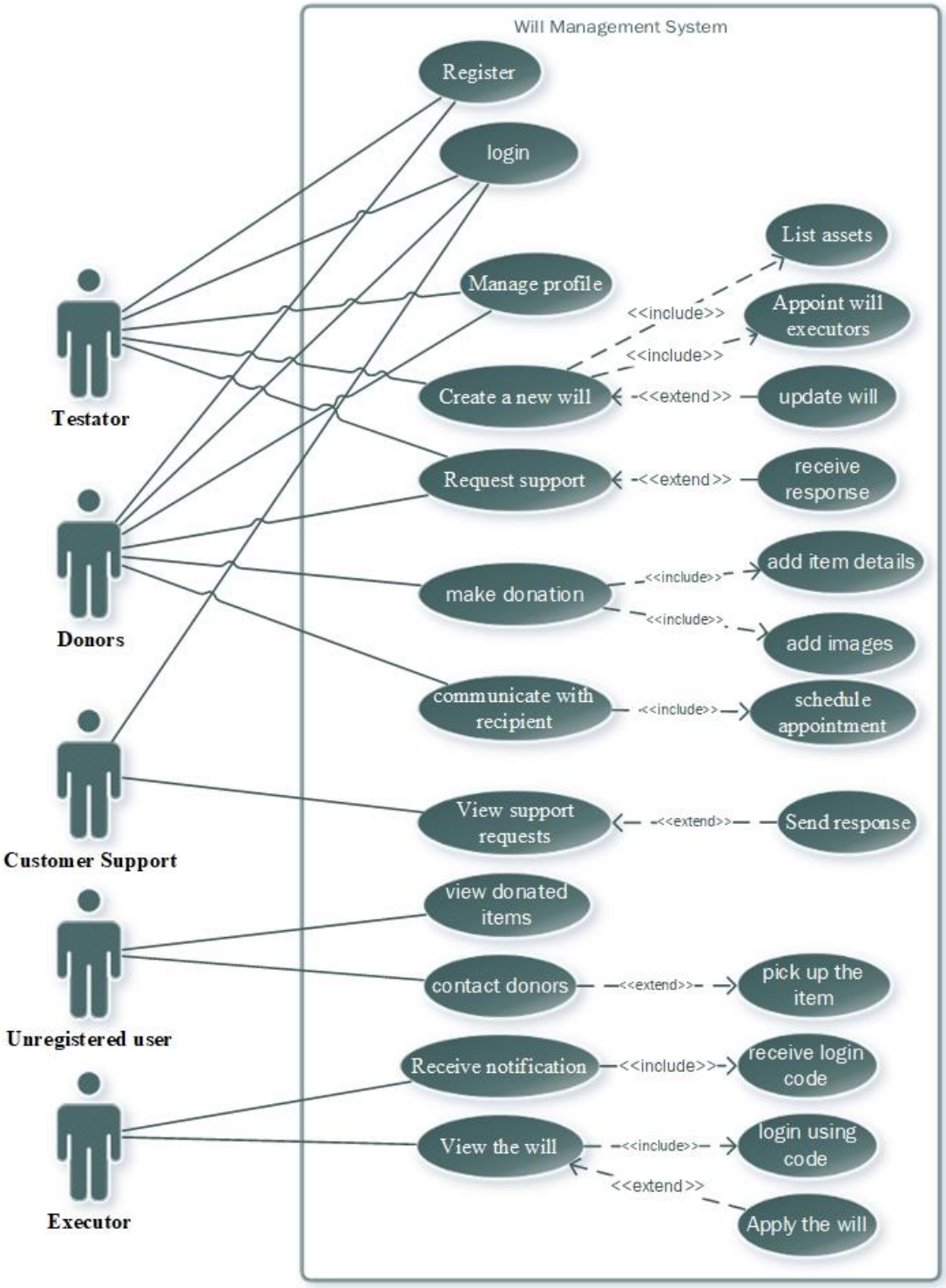

Figure 3.1 Use case diagram.

3.5 Class Diagram

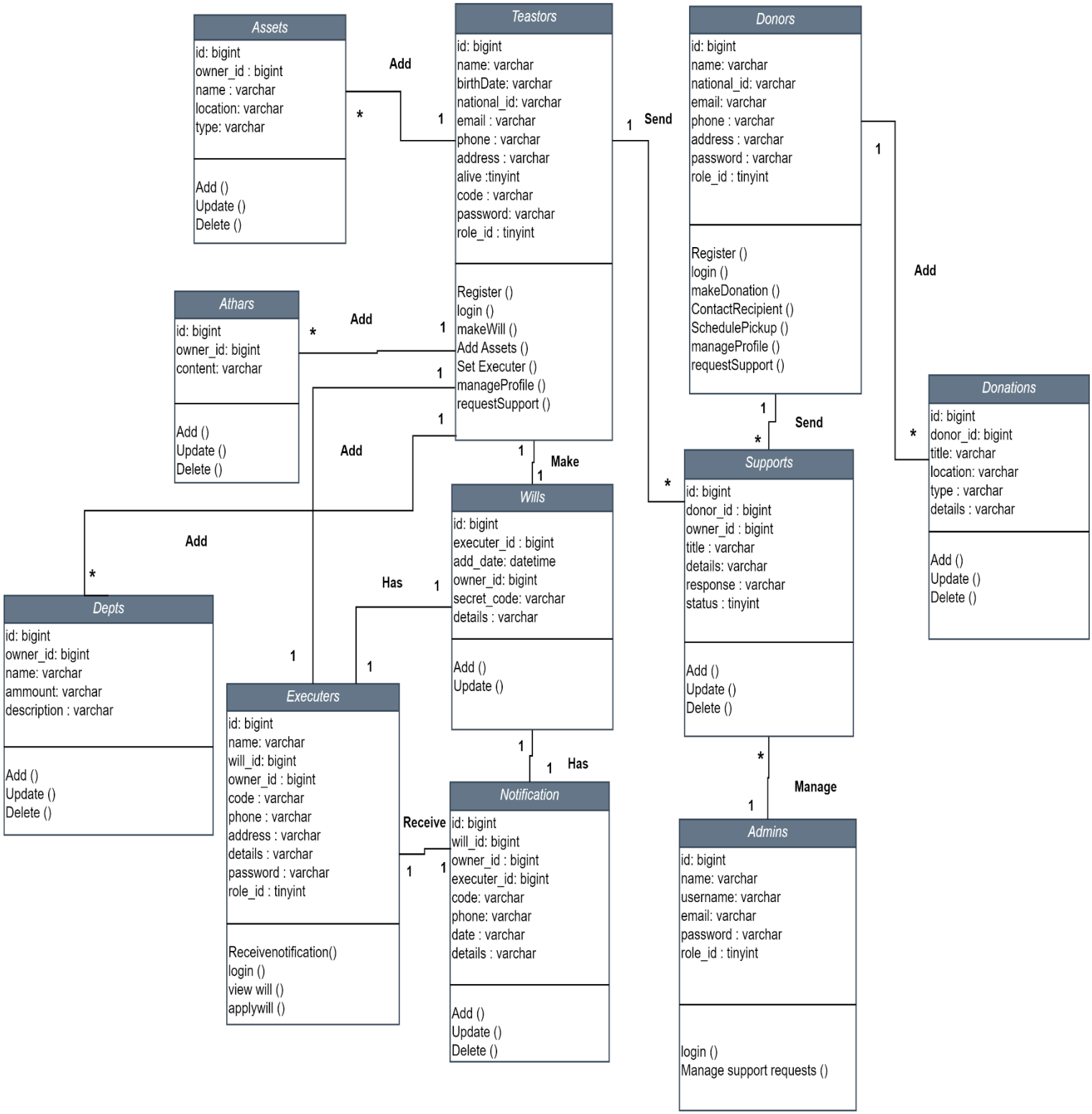

Figure 3.2 Class diagram.

3.6 Sequence Diagram

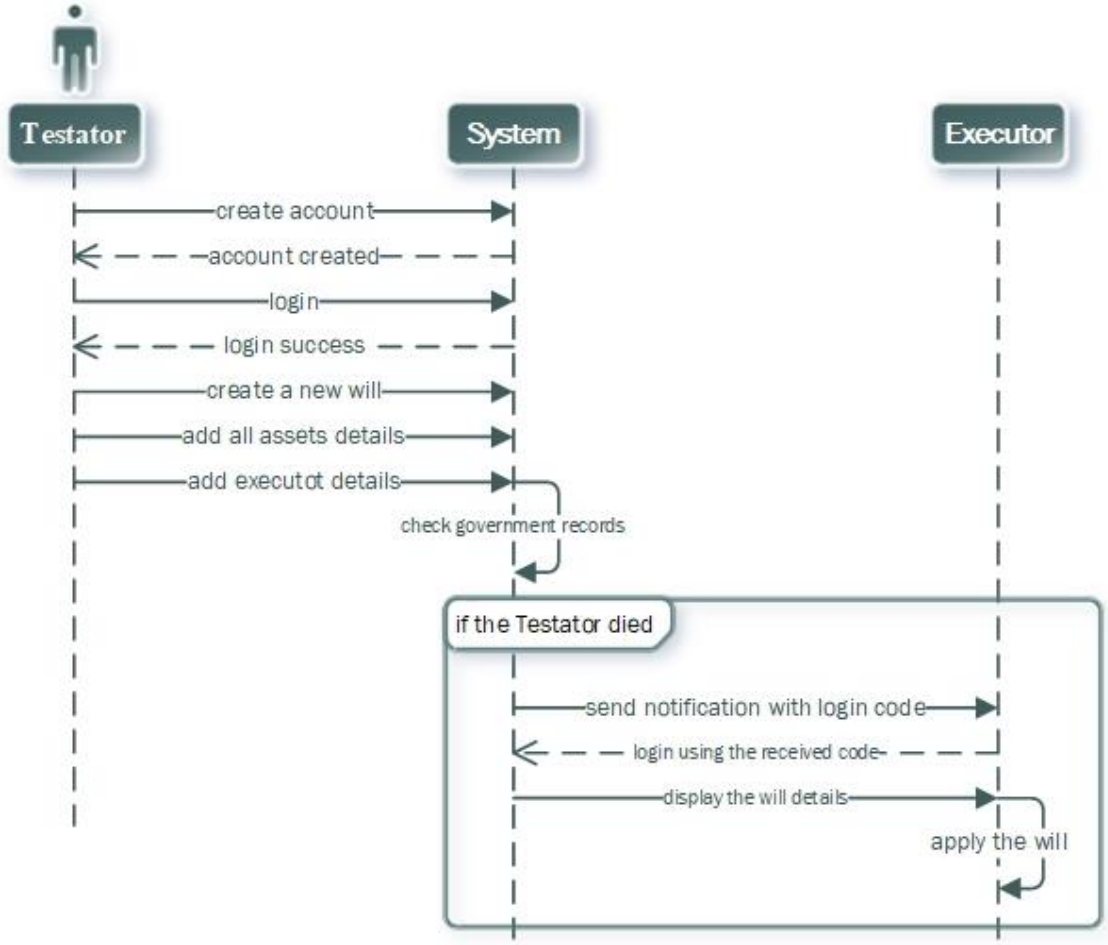

Figure 3.3 Sequence diagram of creating will and applying it.

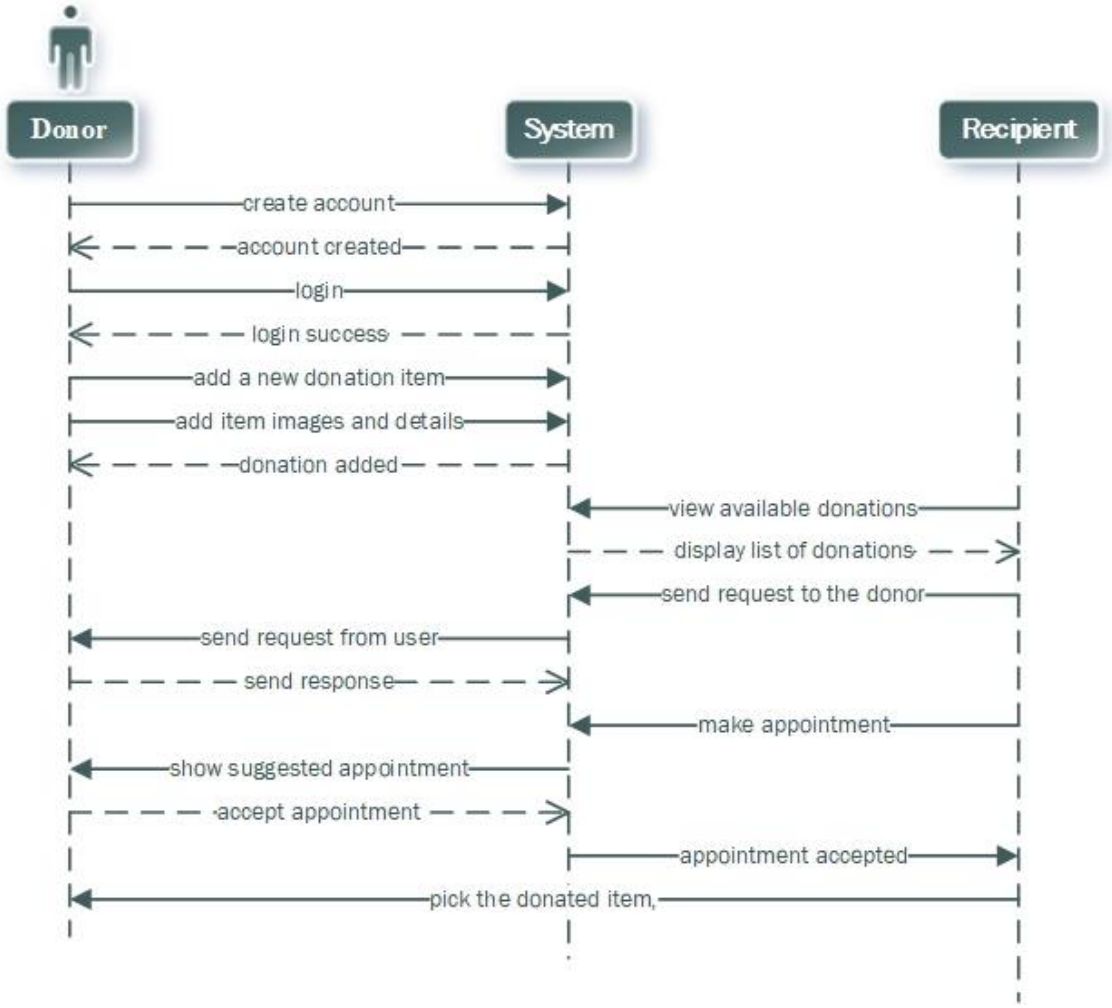

Figure 3.4 Sequence diagram of the donation process.

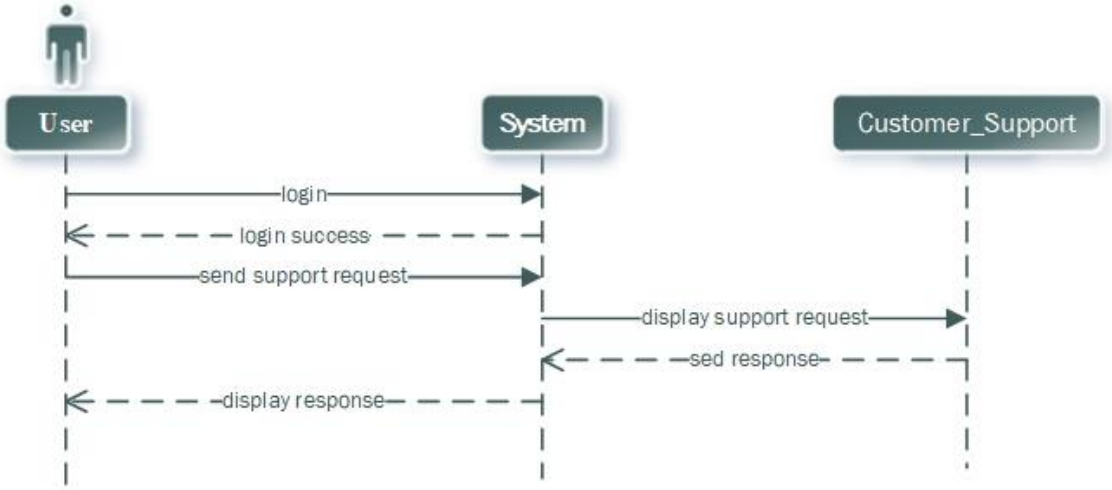

Figure 3.5 Sequence diagram for requesting support.

3.7 Activity Diagram

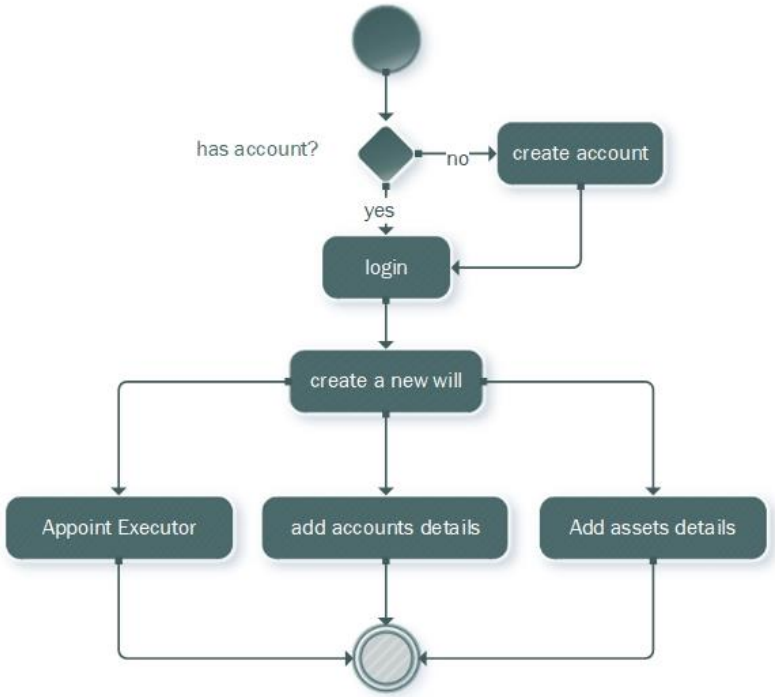

Figure 3.6 Activity diagram of creating the will.

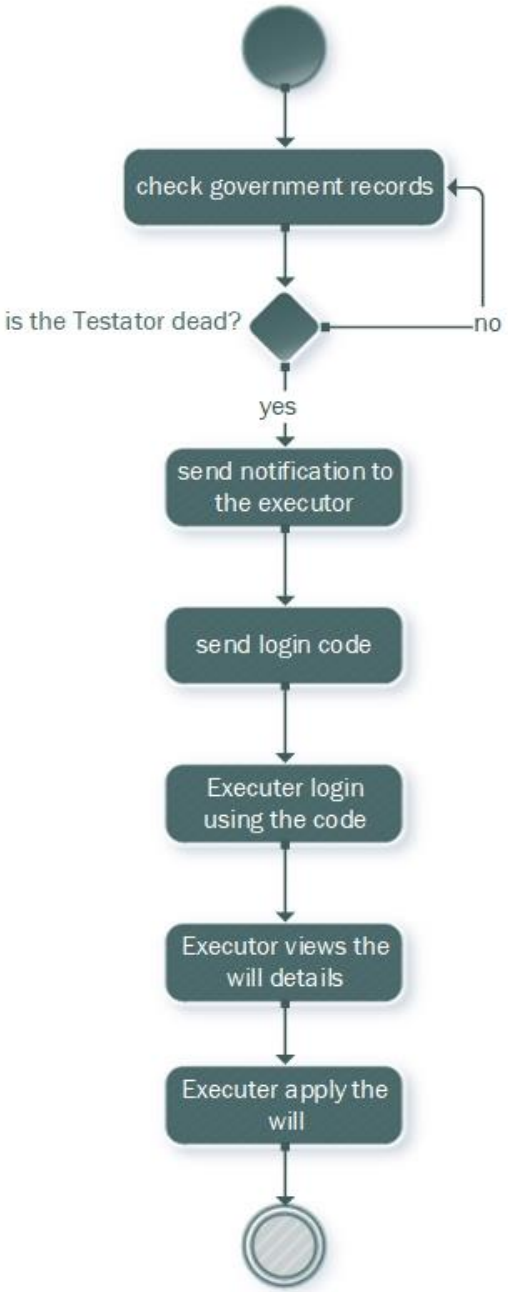

Figure 3.7 Activity diagram for notification and viewing the will.

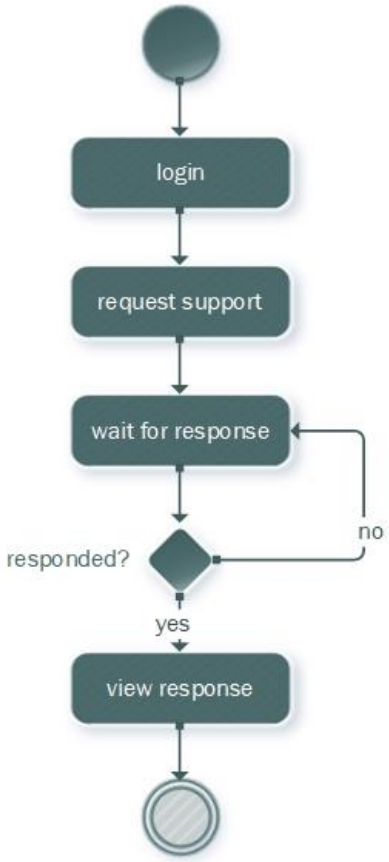

Figure 3.8 Activity diagram for requesting support.

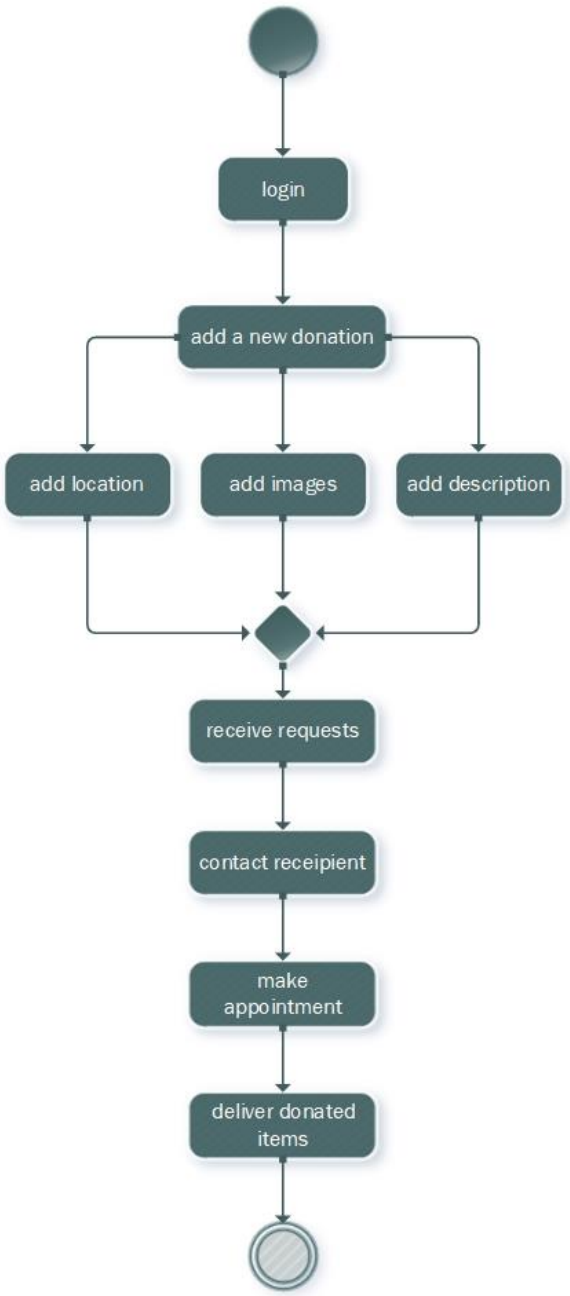

Figure 3.9 Activity diagram for donation process.

Chapter 4 System Design

4.1 State Chart Diagram

A state chart diagram, also known as a state machine diagram or Harel statechart diagram, is a visual representation of the behavior of a system in response to events and internal conditions. It helps model the dynamic aspects of a system by showing how it transitions between different states and how it reacts to various inputs. Components of a State Chart Diagram:

- States: These represent the different "modes" or situations in which a system can exist. They are represented by rectangles with rounded corners.
- Events: These represent external or internal stimuli that can trigger a transition between states. They are represented by arrows labeled with the event name.
- Transitions: These represent the changes from one state to another in response to an event. They are represented by arrows labeled with the event name and the guard condition (optional).
- Initial state: This is the state where the system starts when it is first activated. It is represented by a filled circle inside a state rectangle.
- Final state: This is the state that the system reaches when it completes its execution. It is represented by a circle with a cross inside.

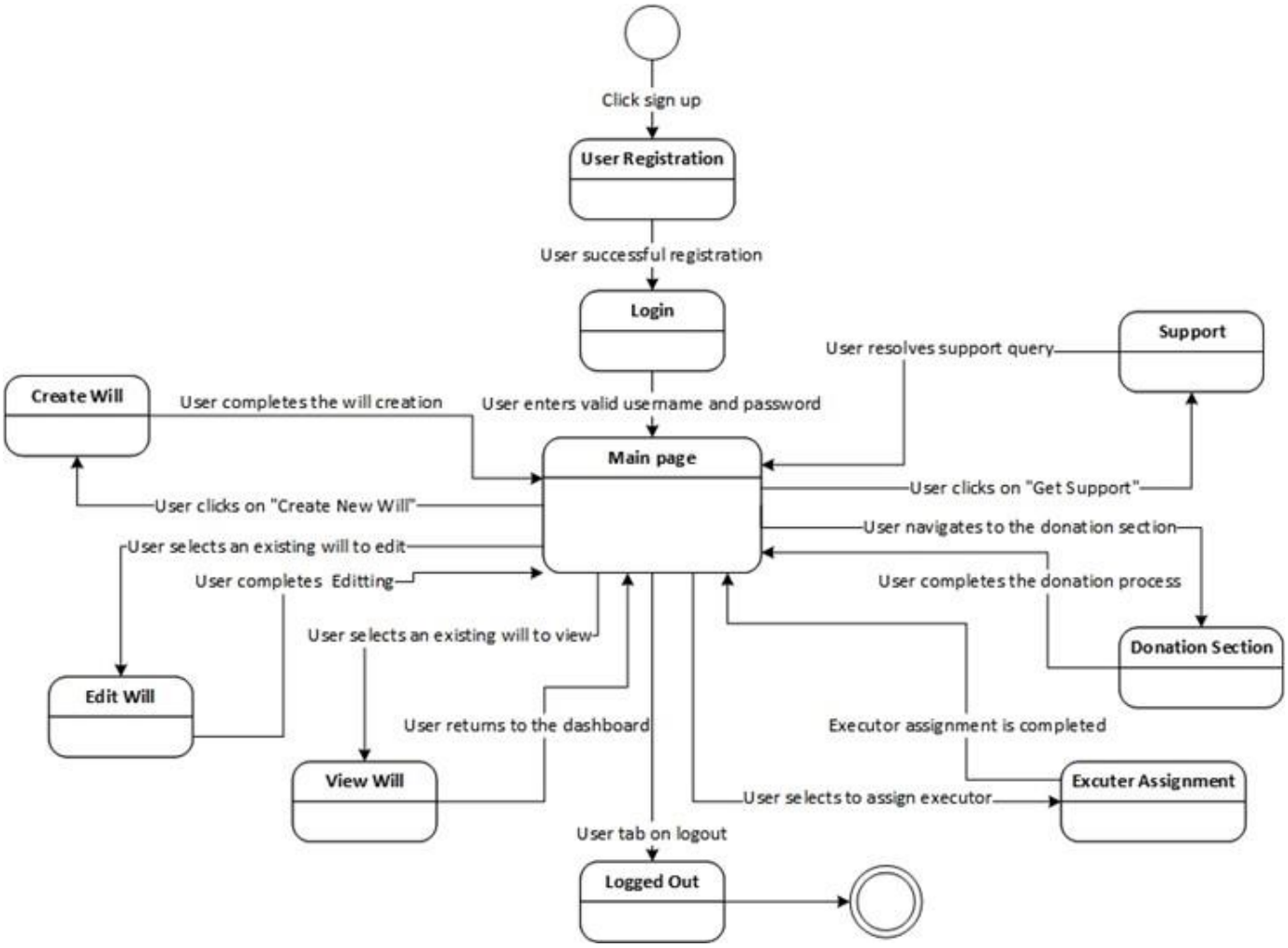

Figure 4.1 State chart diagram.

4.2 ER Diagram

An ER diagram (Entity-Relationship Diagram) is a data modeling tool that shows the relationships between different entities in a database. It is used to identify the different types of entities, their attributes, and the relationships between them. ER diagrams are typically created during the design phase of database development to help define the schema of the database. The key components of an ER diagram:

- Entities: Entities represent the real-world objects that are to be stored in the database. They are typically represented by rectangles.

- Attributes: Attributes represent the properties of an entity. They are typically represented by ovals connected to the entity rectangle.
- Relationships: Relationships show how entities are connected. There are three types of relationships:
    - One-to-one: A one-to-one relationship shows that one entity can be associated with at most one other entity.
    - One-to-many: A one-to-many relationship shows that one entity can be associated with many other entities.
    - Many-to-many: A many-to-many relationship shows that any entity of the first side of the relation can be associated with many entities from the second side. And any entity of the second side of the relation can be associated with many entities from the first side

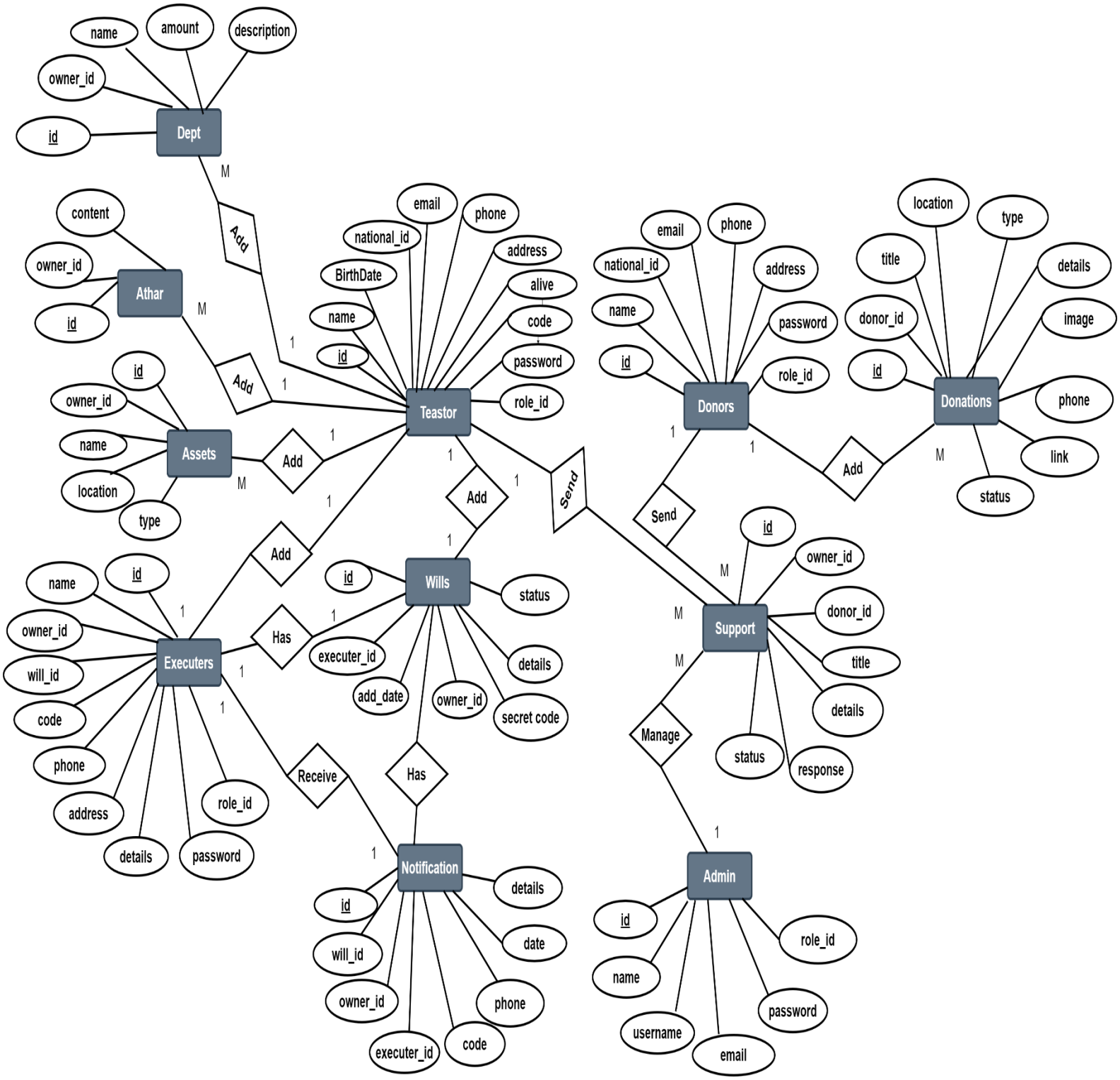

Figure 4.2 ER Diagram

4.3 Database Tables

This section contains the tables of the database, it shows the attributes in each table, data types, keys and others.

Table 4.1 Testator Table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	name	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
3	birthDate	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
4	national_id	varchar(255)	utf8mb4_unicode_ci		No	None		
5	email	varchar(255)	utf8mb4_unicode_ci		No	None		
6	phone	varchar(255)	utf8mb4_unicode_ci		No	None		
7	address	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
8	alive	tinyint(4)			Yes	1		
9	code	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
10	password	varchar(255)	utf8mb4_unicode_ci		No	None		
11	role_id	tinyint(4)			No	2		
12	created_at	timestamp			Yes	NULL		
13	updated_at	timestamp			Yes	NULL		

Table 4.2 Executer table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	name	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
3	will_id	bigint(20)		UNSIGNED	Yes	NULL		
4	owner_id	bigint(20)		UNSIGNED	Yes	NULL		
5	code	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
6	phone	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
7	address	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
8	details	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
9	password	varchar(255)	utf8mb4_unicode_ci		No	None		
10	role_id	tinyint(4)			No	3		
11	created_at	timestamp			Yes	NULL		
12	updated_at	timestamp			Yes	NULL		

Table 4.3 will table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	executer_id	bigint(20)			Yes	NULL		
3	add_date	datetime			Yes	NULL		
4	owner_id	bigint(20)		UNSIGNED	Yes	NULL		
5	secret_code	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
6	details	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
7	status	tinyint(4)			Yes	0		
8	created_at	timestamp			Yes	NULL		
9	updated_at	timestamp			Yes	NULL		

Table 4.4 Notification table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	will_id	bigint(20)		UNSIGNED	Yes	NULL		
3	owner_id	bigint(20)		UNSIGNED	Yes	NULL		
4	executer_id	bigint(20)		UNSIGNED	Yes	NULL		
5	code	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
6	phone	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
7	date	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
8	details	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
9	created_at	timestamp			Yes	NULL		
10	updated_at	timestamp			Yes	NULL		

Table 4.5 the assets table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	owner_id	bigint(20)		UNSIGNED	Yes	NULL		
3	name	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
4	location	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
5	type	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
6	created_at	timestamp			Yes	NULL		
7	updated_at	timestamp			Yes	NULL		

Table 4.6 Donor table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	name	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
3	national_id	varchar(255)	utf8mb4_unicode_ci		No	None		
4	email	varchar(255)	utf8mb4_unicode_ci		No	None		
5	phone	varchar(255)	utf8mb4_unicode_ci		No	None		
6	address	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
7	password	varchar(255)	utf8mb4_unicode_ci		No	None		
8	role_id	tinyint(4)			No	4		
9	created_at	timestamp			Yes	NULL		
10	updated_at	timestamp			Yes	NULL		

Table 4.7 Donation table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	donor_id	bigint(20)		UNSIGNED	Yes	NULL		
3	title	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
4	location	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
5	type	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
6	details	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
7	image	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
8	phone	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
9	link	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
10	status	tinyint(4)			Yes	0		
11	created_at	timestamp			Yes	NULL		
12	updated_at	timestamp			Yes	NULL		

Table 4.8 Dept table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	owner_id	bigint(20)		UNSIGNED	Yes	NULL		
3	name	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
4	ammount	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
5	description	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
6	created_at	timestamp			Yes	NULL		
7	updated_at	timestamp			Yes	NULL		

Table 4.9 Athar table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	owner_id	bigint(20)		UNSIGNED	Yes	NULL		
3	content	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
4	created_at	timestamp			Yes	NULL		
5	updated_at	timestamp			Yes	NULL		

Table 4.10 Support table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT
2	donor_id	bigint(20)		UNSIGNED	Yes	NULL		
3	owner_id	bigint(20)		UNSIGNED	Yes	NULL		
4	title	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
5	details	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
6	response	varchar(255)	utf8mb4_unicode_ci		Yes	NULL		
7	status	tinyint(4)			Yes	NULL		
8	created_at	timestamp			Yes	NULL		
9	updated_at	timestamp			Yes	NULL		

Table 4.11 Admins table.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	bigint(20)		UNSIGNED	No	*None*		AUTO_INCREMENT
2	**name**	varchar(255)	utf8mb4_unicode_ci		Yes	*NULL*		
3	**username**	varchar(255)	utf8mb4_unicode_ci		No	*None*		
4	**email**	varchar(255)	utf8mb4_unicode_ci		No	*None*		
5	**password**	varchar(255)	utf8mb4_unicode_ci		No	*None*		
6	**role_id**	tinyint(4)			No	1		
7	**created_at**	timestamp			Yes	*NULL*		
8	**updated_at**	timestamp			Yes	*NULL*		

4.4 Actual Database

The next figure contains the actual database of the system, is shows the tables and the relation between them.

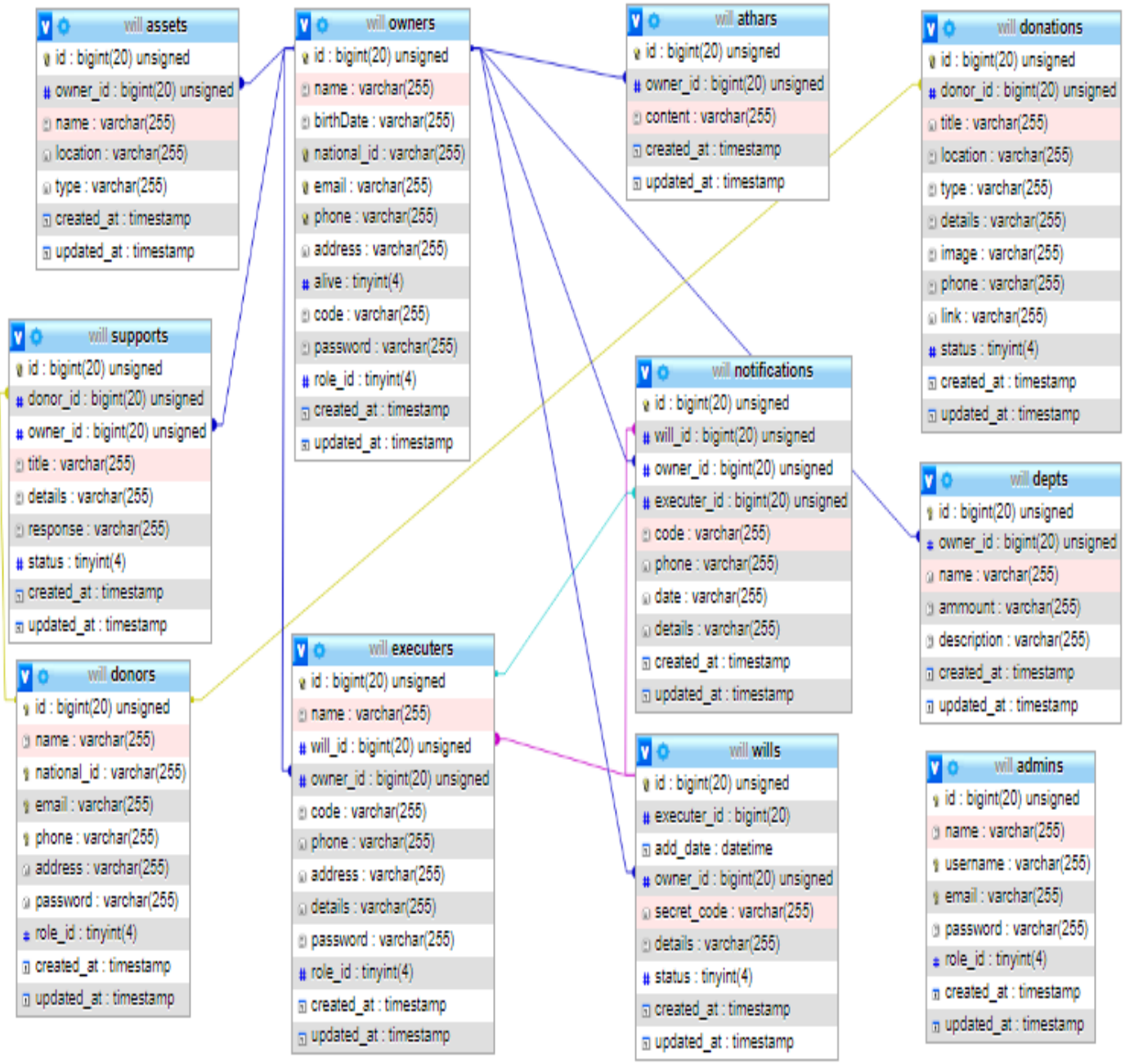

Figure 4.3 the actual database.

4.5 Interfaces prototype

In this section, the user interfaces are displayed to show the expected pages for the application.

Figure 4.4 main page.

Figure 4.5 creating a new account.

Figure 4.6 Open will page.

Figure 4.7 Athar page.

Figure 4.8 will creator page.

Figure 4.9 Donor page.

Figure 4.10 creating will page.

Figure 4.11 Adding assets page.

Figure 4.12 Assign will executer page.

Figure 4.13 Add a new donation.

Figure 4.14 donation list page.

Figure 4.15 Donation details.

Chapter 5 System Implementation

In this section, the most important part of the project will be presented, which is implementing the project, displaying the interfaces that users will interact with and the codes that were used for each interface. The most important tools and techniques that were used to program the application will also be presented.

5.1 Implementation Tools

This project is an electronic application dedicated to Android systems. The Java language and the Laravel framework were used to program the application. Below, a brief overview of the programs and languages that were used to implement the project will be provided.

5.1.1 Java

Java is a programming language that was first released by Sun Microsystems in 1995 and was designed to have the same look and feel as C++, but it is easier to use than C++ and uses an object-oriented programming model [6].

Java can be used to create entire applications that can run on a single computer or distributed among servers and clients in a network. It can also be used to build a module or applet (a simply designed small application) for use as part of a web page.

5.1.2 Android Studio

Android Studio, launched by Google, is a distinctive and unique program for extracting language files and classes and detecting the smallest function, whether encrypted or regular, in APK formats. It is also used in creating and programming applications for the Android system. It relies heavily on the Java language in order to create an application on it [7] .

What makes the Android Studio program distinct from other similar programs is that it provides many features that facilitate the work of the programmer, the most important of which is the program interface, which facilitates the programming process for the user, as

even a beginner can use it and create his first application, and it also provides the ability to write codes in an easy way without errors, as it allows the code that you write to be completed and saved automatically. It also allows the possibility of testing applications on various types of screens and devices via the google cloud service for developers. In addition to this, it supports google services such as google maps, google docs, and provides translation. For more than one language if the programmer needs multiple languages in the application, in addition to many other features.

5.1.3 Laravel

The Laravel framework is one of the most important, most famous, and powerful PHP frameworks. Laravel has been able to prove its worth and occupy leading positions among web application development systems due to its features, the most notable of which is providing many ready-made software functions that make web development easier and faster, and its special interest. By ensuring the security of websites and protecting them from security vulnerabilities. Laravel is an open source PHP-based framework for web application development, developed by Taylor Otwell in 2011 to enable developers to create full-featured web applications easily and quickly by providing the most common tasks and functions they need in web projects [8].

The Laravel framework is characterized by an organized and easy-to-maintain code that keeps programming projects organized through its use of the MVC architecture, which is an abbreviation of three words:

- Model, which means application data, interacts directly with your database and retrieves information from it.
- View, which means the application interface, displays the pages that the user interacts with directly.
- The Controller is the link between the display and the form. It receives user requests, retrieves the required data from the form, processes it, and sends it to the display pages.

5.1.4 Visual Studio Code

Visual Studio code is considered a free and open-source program that can be downloaded on various operating systems with a size not exceeding 100MB. It is also known as VSCode and is a working environment and code editor (IDE AND Code Editor) designed by Microsoft. Despite its small size, it is considered one of the most important script editors ever used. This program appeared after many versions of code editors, which proved unable to keep up with different and modern programming languages such as VI Editor, and Sublime Text Editor. This program offers the ability to provide the necessary rules for writing various modern programming languages by downloading and including offices and packages related to the required language. It has gained huge popularity recently, due to its ability to include libraries and packages for mobile-oriented programming languages such as Flutter. Thus, he was able to display various mobile device emulators without the need to use Android Studio, which was a huge burden on various devices due to its size and power consumption [9].

5.1.5 Xampp

XAMPP is one of the programs for converting a personal computer into a local and multi-use server. As it belongs to the open-source software community, it also works on computers that operate according to Mac, Windows, and Linux systems. It was developed by Apache to make all its features available to everyone for free. It also includes all the tools needed to host local websites with all their details and types [10].

5.2 System Screens

Start interface: This interface appears when you first open the application.

Figure 5.1: Start interface.

Login interface: After an interface, this interface appears to the user. If he has previously registered an account in the system, he can enter the email and password to log in to the application, and otherwise, that is, if he does not previously have an account. This interface contains a link that, by clicking on it, he can go to the page. Create an account. This interface also enables the ordinary user to browse donations or go to the will execution page.

Figure 5.2: Login interface.

Account creation interface: Through this interface, the user can create an account in the application by entering the name, national id number, phone number, email, and password, then choosing his type of use of the system, whether he will register as a donor or as a will writer.

Figure 5.3: Account creation interface.

Displaying donations: Through this interface, the browser can display all donations.

Figure 5.4: Displaying donations.

Interface for displaying donation details: Through this interface, the user can view donation details, such as displaying the donor's name, number, and location.

Figure 5.5: Interface for displaying donation details.

Will opening interface: Through this interface, the user can open his will by entering the will number and the password that is sent to his email.

Figure 5.6: Will opening interface.

Interface for displaying the text of the will: Through this interface, the executor of the will can display the text of the will.

Figure 5.7: Interface for displaying the text of the will.

The main interface for the will writer: Through this interface, the will writer can obtain services from the system, where he can move to the interface for creating a will or adding a valid effect, debts, or property. He can also move to the technical support interface.

Figure 5.8: The main interface for the will writer.

Profile interface: Through this interface, the user can view his personal profile and through it he can log out of the system.

Figure 5.9: Profile interface.

Will creation interface: Through this interface, the will writer can create a serious will, enter its details, and save it in the system.

Figure 5.10: Will creation interface.

Interface for appointing the executor of the will: In this interface, the writer of the will enters the data of the executor of the will, such as his name, email, address, phone number, and the secret number for the will.

Figure 5.11: Interface for appointing the executor of the will.

Interface for adding items: Through this interface, the will writer can enter items and his property.

Figure 5.12: Interface for adding items.

Adding a good impact interface: Through this interface, the will writer can recommend a good deed after his death.

Figure 5.13: Adding a good impact interface.

Add debt interface: If the will writer has a debt, through this interface he can enter the amount of debt that he must pay.

Figure 5.14: Add debt interface.

Technical support interface: Through this interface, the will writer can request technical support in the event of any malfunction.

Figure 5.15: Technical support interface.

The main interface for the donor: Through this interface, the donor can obtain his services from the system, where he can view the donations he has added or create new donations. He can also move to the technical support interface.

Figure 5.16: The main interface for the donor.

Interface for displaying previously added donations: The donor can view all the donations that he has added to the system through this interface.

Figure 5.17: Interface for displaying previously added donations.

Add a new donation interface: Through this interface, the donor can enter a new donation into the system.

Figure 5.18: Add a new donation interface.

5.3 CODE SAMPLES

5.3.1 User registration

- **THE XML CODE**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".RegisterActivity">

    <androidx.constraintlayout.widget.ConstraintLayout
        android:layout_width="match_parent"
        android:layout_height="690dp"
        android:background="@drawable/layout_bg"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
```

```
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="1.0" >

<TextView
    android:id="@+id/textView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:fontFamily="@font/arapey"
    android:text="@string/register_new_text"
    android:textColor="@color/white"
    android:textSize="30dp"
    android:textStyle="bold"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.025" />

<EditText
    android:id="@+id/username"
    android:layout_width="350dp"
    android:layout_height="60dp"
    android:background="@drawable/text_bg"
    android:drawableRight="@drawable/baseline_person_2_24"
    android:drawablePadding="10dp"
    android:hint="@string/name4"
    android:padding="20dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.491"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/textView2"
    app:layout_constraintVertical_bias="0.034" />

<EditText
    android:id="@+id/identity"
    android:layout_width="350dp"
    android:layout_height="60dp"
    android:background="@drawable/text_bg"
    android:drawableRight="@drawable/baseline_dataset_24"
    android:drawablePadding="10dp"
    android:hint="@string/identity"
    android:padding="20dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.491"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/username"
    app:layout_constraintVertical_bias="0.046" />

<EditText
    android:id="@+id/phone"
    android:layout_width="350dp"
    android:layout_height="60dp"
    android:background="@drawable/text_bg"
    android:drawableRight="@drawable/phone"
```

```
        android:drawablePadding="10dp"
        android:hint="@string/phone_number"
        android:padding="20dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.508"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/identity"
        app:layout_constraintVertical_bias="0.046" />

    <EditText
        android:id="@+id/email"
        android:layout_width="350dp"
        android:layout_height="60dp"
        android:background="@drawable/text_bg"
        android:drawableRight="@drawable/baseline_markunread_24"
        android:drawablePadding="10dp"
        android:hint="@string/email"
        android:padding="20dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.491"
        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintTop_toBottomOf="@+id/phone"
        app:layout_constraintVertical_bias="0.051" />

    <EditText
        android:id="@+id/password"
        android:layout_width="350dp"
        android:layout_height="60dp"
        android:inputType="textPassword"
        android:background="@drawable/text_bg"
        android:drawableRight="@drawable/baseline_mail_lock_24"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.491"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/email"
        app:layout_constraintVertical_bias="0.094"
        android:padding="20dp"
        android:drawablePadding="10dp"
        android:hint="@string/password"
        />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/user_type_text"
        android:textColor="@color/white"
        android:textSize="20dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.883"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/password"
```

```
            app:layout_constraintVertical_bias="0.088" />

        <RadioGroup
            android:id="@+id/idradiogroup"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginLeft="10dp"
            android:layout_marginRight="10dp"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintHorizontal_bias="0.0"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@+id/textView3"
            app:layout_constraintVertical_bias="0.284">

            <RadioButton
                android:id="@+id/male"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layoutDirection="rtl"
                android:text="@string/will_writer_text"
                android:textColor="@color/white" />

            <RadioButton
                android:id="@+id/feminine"
                android:layoutDirection="rtl"
                android:textColor="@color/white"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="@string/donor_text" />
        </RadioGroup>

        <androidx.appcompat.widget.AppCompatButton
            android:id="@+id/reg"
            android:layout_width="180dp"
            android:layout_height="50dp"
            android:background="@drawable/text_bg"
            android:text="@string/regText2"
            android:textColor="@color/text_color"
          android:textAllCaps="false"
            android:layout_marginBottom="5dp"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintHorizontal_bias="0.558"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@+id/idradiogroup"
            app:layout_constraintVertical_bias="0.0" />
    </androidx.constraintlayout.widget.ConstraintLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

- **THE JAVA CODE**

```
public class RegisterActivity extends AppCompatActivity {
```

```
    Button reg;
    private EditText username,identity,phone,email,password;
    int role_id;
    private RadioGroup radioGrp;
    String type;
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_register);
        reg=findViewById(R.id.reg);
        username=findViewById(R.id.username);
        identity=findViewById(R.id.identity);
        phone=findViewById(R.id.phone);
        email=findViewById(R.id.email);
        password=findViewById(R.id.password);
        radioGrp=findViewById(R.id.idradiogroup);

        username.setInputType(InputType.TYPE_CLASS_TEXT |
InputType.TYPE_TEXT_VARIATION_NORMAL);
        radioGrp.setOnCheckedChangeListener(new
RadioGroup.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(RadioGroup group, int
checkedId) {
                RadioButton radioButton = findViewById(checkedId);
                if (radioButton == findViewById(R.id.male)) {
                    type = "الوصية كاتب";
                    role_id=2;

                } else if (radioButton == findViewById(R.id.feminine))
{
                    type = "متبرع";
                    role_id=4;
                }
            }
        });


        reg.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                registerUser();
            }
        });
    }
    private void registerUser(){

        final String myName = username.getText().toString().trim();
        Charset.forName("UTF-8").encode(myName);
        Log.d(RegisterActivity.class.getName(), "onResponse: "+myName);
        final String myEmail = email.getText().toString().trim();
        final String myPhone = phone.getText().toString().trim();
        final String myPassword = password.getText().toString().trim();
        final String myIdentity =identity.getText().toString().trim();
```

```
        if (!Patterns.EMAIL_ADDRESS.matcher(myEmail).matches()) {
            email.setError("Enter you email Correctly");
            email.requestFocus();
            return;
        }
        if (myPassword.length()<=8){
            password.setError("Enter your password must be more than 8
charchter");
            password.requestFocus();
            return;
        }
        if (myIdentity.length()>10){
            identity.setError("Your identity number must not be more
than 10 charchter");
            identity.requestFocus();
            return;
        }
        if (myPhone.length()>10 ){
            phone.setError("Enter your Mobile number Correctly");
            phone.requestFocus();
            return;
        }


        if (!myPhone.substring(0, 2).equals("05")) {
            phone.setError("Enter your Mobile number correctly");
            phone.requestFocus();
            return;
        }
        if(TextUtils.isEmpty(myName)){
            username.setError("Enter you name please");
            username.requestFocus();
            return;
        }
        if(TextUtils.isEmpty(myIdentity)){
           identity.setError("Enter you Identity number please");
            identity.requestFocus();
            return;
        }
        if(TextUtils.isEmpty(myEmail)){
            email.setError("Enter you email please");
            email.requestFocus();
            return;
        }
        if(TextUtils.isEmpty(myPhone)){
            phone.setError("Enter you phone please");
            phone.requestFocus();
            return;
        }
        if(TextUtils.isEmpty(myPassword)){
            password.setError("Enter you password please");
            password.requestFocus();
            return;
        }
```

```
        StringRequest stringRequest = new
StringRequest(Request.Method.POST, URLs.URL_REGISTER,
                new Response.Listener<String>() {
                    @Override
                    public void onResponse(String response) {

                        try {
                            JSONObject jsonObject = new
JSONObject(response);
                            Log.d(RegisterActivity.class.getName(),
"onResponse: "+response);
                            if (jsonObject.getInt("status")!=-1){

                                Toast.makeText(RegisterActivity.this,

jsonObject.getString("message"), Toast.LENGTH_SHORT).show();
                                JSONObject userObject=
jsonObject.getJSONObject("user");
                                User user = new
User(userObject.getInt("id"));


SessionManagerUser.getInstance(RegisterActivity.this).userLogin(user);
                                if (role_id==2){
                                    Intent intent = new
Intent(RegisterActivity.this, WillCreatorMainActivity.class);
                                    startActivity(intent);
                                } else if (role_id==4) {
                                    Intent intent = new
Intent(RegisterActivity.this, DonorMainActivity.class);
                                    startActivity(intent);
                                }

                            } else {

Toast.makeText(RegisterActivity.this,"Register
Failed",Toast.LENGTH_SHORT).show();

                            }
                        } catch (JSONException e) {
                            throw new RuntimeException(e);
                        }

                    }
                },
                new Response.ErrorListener() {
                    @Override
                    public void onErrorResponse(VolleyError error) {


Toast.makeText(RegisterActivity.this,error.toString(),Toast.LENGTH_SHOR
T).show();
                    }

                }
        ) {
```

```
            @Override
            protected Map<String, String> getParams() throws
AuthFailureError {
                Map<String, String> params = new HashMap<>();
                params.put("role_id",String.valueOf(role_id));
                params.put("name", myName);
                params.put("phone", myPhone);
                params.put("password", myPassword);
                params.put("email", myEmail);
                params.put("national_id", myIdentity);
                return params;
            }

        }
                ;

VolleySingleton.getInstance(RegisterActivity.this).addToRequestQueue(st
ringRequest);
    }
}
```

5.3.2 Will Creator Main Activity

- **THE XML CODE**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".WillCreatorMainActivity"
    android:background="@color/text_color">

    <TextView
        android:id="@+id/textView14"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/hello"
        android:textColor="@color/white"
        android:textSize="30dp"
        android:textStyle="bold"

app:layout_constraintBottom_toTopOf="@+id/constraintLayout3"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.916"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.609" />

    <ImageView
        android:id="@+id/profielc"
```

```
        android:layout_width="54dp"
        android:layout_height="46dp"
        android:src="@drawable/person"

app:layout_constraintBottom_toTopOf="@+id/constraintLayout3"
        app:layout_constraintEnd_toStartOf="@+id/textView14"
        app:layout_constraintHorizontal_bias="0.138"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.444" />


    <androidx.constraintlayout.widget.ConstraintLayout
        android:id="@+id/constraintLayout3"
        android:layout_width="match_parent"
        android:layout_height="650dp"
        android:background="@drawable/layout_bg_wh"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="1.0">

        <LinearLayout
            android:id="@+id/linearLayout"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="40dp"
            android:gravity="center_horizontal"
            android:orientation="horizontal"
            android:weightSum="2"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toTopOf="parent">

            <androidx.cardview.widget.CardView
                android:id="@+id/contributeCard"
                android:layout_width="128dp"
                android:layout_height="128dp"
                android:layout_margin="8dp"
                android:layout_marginEnd="16dp"
                android:clickable="true"
                android:elevation="12dp"
                android:focusable="true"

android:foreground="?android:attr/selectableItemBackground"
                app:cardCornerRadius="8dp">


<androidx.constraintlayout.widget.ConstraintLayout
                    android:id="@+id/adding_assest"
                    android:layout_width="match_parent"
                    android:layout_height="match_parent"
                    android:background="@color/text_color">

                    <ImageView
```

```
                        android:id="@+id/image_view1"
                        android:layout_width="0dp"
                        android:layout_height="50dp"
                        android:src="@drawable/assest"

app:layout_constraintBottom_toTopOf="@+id/textView1"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"
                        app:layout_constraintTop_toTopOf="parent"
/>

                    <TextView
                        android:id="@+id/textView1"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:gravity="center"
                        android:text="@string/addAssest"

                        android:textColor="@color/white"
                        android:textSize="15sp"
                        android:textStyle="bold"

app:layout_constraintBottom_toBottomOf="parent"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/image_view1" />


</androidx.constraintlayout.widget.ConstraintLayout>

            </androidx.cardview.widget.CardView>

            <androidx.cardview.widget.CardView
                android:id="@+id/interestsCard"
                android:layout_width="128dp"
                android:layout_height="128dp"
                android:layout_margin="8dp"
                android:layout_marginStart="16dp"
                android:clickable="true"
                android:elevation="12dp"
                android:focusable="true"

android:foreground="?android:attr/selectableItemBackground"
                app:cardCornerRadius="8dp">


<androidx.constraintlayout.widget.ConstraintLayout
                    android:id="@+id/create_will"
                    android:layout_width="match_parent"
                    android:layout_height="match_parent"
```

```
                    android:background="@color/text_color">

                    <ImageView
                        android:id="@+id/image_view22"
                        android:layout_width="0dp"
                        android:layout_height="60dp"
                        android:src="@drawable/will_create"

app:layout_constraintBottom_toTopOf="@+id/textView22"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"
                        app:layout_constraintTop_toTopOf="parent"
/>

                    <TextView
                        android:id="@+id/textView22"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:gravity="center"
                        android:text="@string/create_your_will"

                        android:textColor="@color/white"
                        android:textSize="15sp"
                        android:textStyle="bold"

app:layout_constraintBottom_toBottomOf="parent"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/image_view22" />


</androidx.constraintlayout.widget.ConstraintLayout>

            </androidx.cardview.widget.CardView>
        </LinearLayout>


        <LinearLayout
            android:id="@+id/linearLayout3"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center_horizontal"
            android:orientation="horizontal"
            android:weightSum="2"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/linearLayout">

            <androidx.cardview.widget.CardView
```

```
                android:id="@+id/settingsCard"
                android:layout_width="128dp"
                android:layout_height="128dp"
                android:layout_margin="8dp"
                android:layout_marginStart="16dp"
                android:clickable="true"
                android:elevation="12dp"
                android:focusable="true"

android:foreground="?android:attr/selectableItemBackground"
                app:cardCornerRadius="8dp">


<androidx.constraintlayout.widget.ConstraintLayout
                    android:id="@+id/debts_layoutD"
                    android:layout_width="match_parent"
                    android:layout_height="match_parent"
                    android:background="@color/text_color">

                    <ImageView
                        android:id="@+id/dProfiel"
                        android:layout_width="0dp"
                        android:layout_height="60dp"
                        android:src="@drawable/people_d"

app:layout_constraintBottom_toTopOf="@+id/textView5"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"
                        app:layout_constraintTop_toTopOf="parent"
/>

                    <TextView
                        android:id="@+id/textView5"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:gravity="center"
                        android:text="@string/debts_text"
                        android:textColor="@color/white"
                        android:textSize="15sp"
                        android:textStyle="bold"

app:layout_constraintBottom_toBottomOf="parent"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.0"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/dProfiel" />


</androidx.constraintlayout.widget.ConstraintLayout>

            </androidx.cardview.widget.CardView>
```

```
            <androidx.cardview.widget.CardView
                android:id="@+id/settingsCard1"
                android:layout_width="128dp"
                android:layout_height="128dp"
                android:layout_margin="8dp"
                android:layout_marginStart="16dp"
                android:clickable="true"
                android:elevation="12dp"
                android:focusable="true"

android:foreground="?android:attr/selectableItemBackground"
                app:cardCornerRadius="8dp">


<androidx.constraintlayout.widget.ConstraintLayout
                    android:id="@+id/athar_layout"
                    android:layout_width="match_parent"
                    android:layout_height="match_parent"
                    android:background="@color/text_color">

                    <ImageView
                        android:id="@+id/image_view"
                        android:layout_width="0dp"
                        android:layout_height="60dp"
                        android:src="@drawable/athar"

app:layout_constraintBottom_toTopOf="@+id/textView4"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"
                        app:layout_constraintTop_toTopOf="parent"
/>

                    <TextView
                        android:id="@+id/textView4"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:gravity="center"
                        android:text="@string/good_effect_text"
                        android:textColor="@color/white"
                        android:textSize="15sp"
                        android:textStyle="bold"

app:layout_constraintBottom_toBottomOf="parent"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/image_view" />


</androidx.constraintlayout.widget.ConstraintLayout>
```

```
            </androidx.cardview.widget.CardView>

        </LinearLayout>

        <LinearLayout
            android:id="@+id/linearLayout2"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:gravity="center_horizontal"
            android:orientation="horizontal"
            android:weightSum="2"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/linearLayout3">

            <androidx.cardview.widget.CardView
                android:id="@+id/learnCard"
                android:layout_width="128dp"
                android:layout_height="128dp"
                android:layout_margin="8dp"
                android:layout_marginEnd="16dp"
                android:clickable="true"
                android:elevation="12dp"
                android:focusable="true"

android:foreground="?android:attr/selectableItemBackground"
                app:cardCornerRadius="8dp">


<androidx.constraintlayout.widget.ConstraintLayout
                    android:id="@+id/supportd_layout"
                    android:layout_width="match_parent"
                    android:layout_height="match_parent"
                    android:background="@color/text_color">

                    <ImageView
                        android:id="@+id/image_view21"
                        android:layout_width="0dp"
                        android:layout_height="50dp"
                        android:src="@drawable/support"

app:layout_constraintBottom_toTopOf="@+id/textView21"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"
                        app:layout_constraintTop_toTopOf="parent"
/>

                    <TextView
                        android:id="@+id/textView21"
                        android:layout_width="match_parent"
                        android:layout_height="wrap_content"
                        android:gravity="center"
```

```
                        android:text="@string/technical_support"

                        android:textColor="@color/white"
                        android:textSize="15sp"
                        android:textStyle="bold"

app:layout_constraintBottom_toBottomOf="parent"
                        app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintHorizontal_bias="0.5"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/image_view21" />


</androidx.constraintlayout.widget.ConstraintLayout>

            </androidx.cardview.widget.CardView>


        </LinearLayout>

    </androidx.constraintlayout.widget.ConstraintLayout>

    <TextView
        android:id="@+id/eng"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="35dp"
        android:text="En"
        android:textColor="@color/white"
        android:textSize="20dp"
        android:layout_marginLeft="10dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toStartOf="@+id/textView14"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toEndOf="@+id/profielc"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.037" />
    <TextView
        android:id="@+id/textviewe"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="35dp"
        android:text="|"
        android:textColor="@color/white"
        android:textSize="20dp"
        android:textStyle="bold"
        android:layout_marginLeft="35dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toStartOf="@+id/textView14"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toEndOf="@+id/profielc"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.037" />
    <TextView
```

```
            android:id="@+id/arb"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginBottom="35dp"
            android:text="Ar"
            android:textColor="@color/white"
            android:textSize="20dp"
            android:layout_marginLeft="45dp"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toStartOf="@+id/textView14"
            app:layout_constraintHorizontal_bias="0.0"
            app:layout_constraintStart_toEndOf="@+id/profielc"
            app:layout_constraintTop_toTopOf="parent"
            app:layout_constraintVertical_bias="0.037" />
    </androidx.constraintlayout.widget.ConstraintLayout>
```

- **THE JAVA CODE**

```
public class WillCreatorMainActivity extends AppCompatActivity {

    ConstraintLayout
create_will,adding_assest,athar_layout,debts_layoutD,supportd_layout;
    ImageView profielc;

    TextView en,ar;

    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_will_creator_main);

        create_will=findViewById(R.id.create_will);
        adding_assest=findViewById(R.id.adding_assest);
        profielc=findViewById(R.id.profielc);
        athar_layout=findViewById(R.id.athar_layout);
        debts_layoutD=findViewById(R.id.debts_layoutD);
        supportd_layout=findViewById(R.id.supportd_layout);
        en=findViewById(R.id.eng);
        ar=findViewById(R.id.arb);



        ar.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Locale locale = new Locale("ar");
                Locale.setDefault(locale);
                Configuration config = new Configuration();
                config.locale = locale;
                getResources().updateConfiguration(config,
getResources().getDisplayMetrics());
                recreate();
```

```
            }


        });
        en.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Locale locale = new Locale("en");
                Locale.setDefault(locale);
                Configuration config = new Configuration();
                config.locale = locale;
                getResources().updateConfiguration(config,
getResources().getDisplayMetrics());
                recreate();



            }
        });


        create_will.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new
Intent(WillCreatorMainActivity.this,CreateWillActivity.class);
                startActivity(intent);
            }
        });
        adding_assest.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new
Intent(WillCreatorMainActivity.this, AddingAssetsActivity.class);
                startActivity(intent);
            }
        });
        profielc.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new
Intent(WillCreatorMainActivity.this,WillCProfielActivity.class);
                startActivity(intent);
            }
        });
        athar_layout.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new
Intent(WillCreatorMainActivity.this, WriteAtharActivity.class);
                startActivity(intent);
            }
        });
        debts_layoutD.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new
```

```
Intent(WillCreatorMainActivity.this, DebtsWillCreateorActivity.class);
                startActivity(intent);
            }
        });
        supportd_layout.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new
Intent(WillCreatorMainActivity.this,
WillCreateorSupportListActivity.class);
                startActivity(intent);
            }
        });
    }
```

5.3.3 Create New Donation Activity

- **THE XML CODE**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".CreateNewDonationActivity"
    android:background="@color/text_color"
    >

    <ImageView
        android:layout_width="66dp"
        android:layout_height="51dp"
        android:src="@drawable/addw"

app:layout_constraintBottom_toTopOf="@+id/constraintLayout4"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toEndOf="@+id/textView16"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.795" />

    <TextView
        android:id="@+id/textView16"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/create_donation"
        android:textColor="@color/white"
        android:textSize="25dp"
        android:textStyle="bold"

app:layout_constraintBottom_toTopOf="@+id/constraintLayout4"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.695"
```

```
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.737" />

    <androidx.constraintlayout.widget.ConstraintLayout
        android:id="@+id/constraintLayout4"
        android:layout_width="match_parent"
        android:layout_height="620dp"

        android:background="@drawable/layout_bg_wh"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.855">


        <EditText
            android:id="@+id/e2"
            android:layout_width="350dp"
            android:layout_height="60dp"
            android:background="@drawable/text_bg"
            android:drawablePadding="10dp"
            android:hint="@string/add_title"
            android:padding="10dp"
            android:inputType="text"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintHorizontal_bias="0.491"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toTopOf="parent"
            app:layout_constraintVertical_bias="0.142" />

        <EditText
            android:id="@+id/e3"
            android:layout_width="350dp"
            android:layout_height="60dp"
            android:background="@drawable/text_bg"
            android:drawablePadding="10dp"
            android:hint="@string/add_description"
            android:inputType="text"
            android:padding="10dp"
            app:layout_constraintBottom_toBottomOf="parent"
            app:layout_constraintEnd_toEndOf="parent"
            app:layout_constraintHorizontal_bias="0.491"
            app:layout_constraintStart_toStartOf="parent"
            app:layout_constraintTop_toBottomOf="@+id/e2"
            app:layout_constraintVertical_bias="0.069" />

        <EditText
            android:id="@+id/e5"
            android:layout_width="350dp"
            android:layout_height="60dp"
            android:background="@drawable/text_bg"
            android:drawablePadding="10dp"
            android:hint="@string/add_site"
```

```
        android:inputType="text"
        android:padding="10dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.491"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/e3"
        app:layout_constraintVertical_bias="0.105" />

    <TextView
        android:id="@+id/textView25"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/add_photo"
        android:textColor="@color/black"
        android:textSize="20dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.773"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/e5"
        app:layout_constraintVertical_bias="0.203" />

    <ImageView
        android:layout_width="38dp"
        android:layout_height="23dp"
        android:src="@drawable/add"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.138"
        app:layout_constraintStart_toEndOf="@+id/textView25"
        app:layout_constraintTop_toBottomOf="@+id/e5"
        app:layout_constraintVertical_bias="0.222" />

    <ImageView
        android:id="@+id/image"
        android:layout_width="166dp"
        android:layout_height="101dp"
        android:src="@drawable/image_"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toStartOf="@+id/textView25"
        app:layout_constraintHorizontal_bias="0.476"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/e5"
        app:layout_constraintVertical_bias="0.097" />

    <androidx.constraintlayout.widget.ConstraintLayout
        android:id="@+id/constraintLayout7"
        android:layout_width="match_parent"
        android:layout_height="100dp"
        android:background="@drawable/layout_bg_d"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent">

        <TextView
```

```
                    android:id="@id/done"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="@string/save"
                    android:textColor="@color/white"
                    android:textSize="25dp"
                    android:textStyle="bold"
                    app:layout_constraintBottom_toBottomOf="parent"
                    app:layout_constraintEnd_toEndOf="parent"
                    app:layout_constraintStart_toStartOf="parent"
                    app:layout_constraintTop_toTopOf="parent" />


            </androidx.constraintlayout.widget.ConstraintLayout>
        </androidx.constraintlayout.widget.ConstraintLayout>

    </androidx.constraintlayout.widget.ConstraintLayout>
```

- **THE JAVA CODE**

```
public class CreateNewDonationActivity extends AppCompatActivity {

    private EditText title,details,location;
    private TextView addImg,save;
    private ImageView img;
    int SELECT_PICTURE = 1222;
    Uri fileUri;
    Bitmap b;
    private int id;
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_create_new_donation);

        title=findViewById(R.id.e2);
        details=findViewById(R.id.e3);
        location=findViewById(R.id.e5);
        addImg=findViewById(R.id.textView25);
        img=findViewById(R.id.image);
        save=findViewById(R.id.done);

        addImg.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
               imageChooser();

            }
        });


        save.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
            public void onClick(View v) {
                saveData(b);
            }
        });

    }
    void imageChooser() {


        Intent i = new Intent();
        i.setType("image/*");
        i.setAction(Intent.ACTION_GET_CONTENT);


        startActivityForResult(Intent.createChooser(i, "Select
Picture"), SELECT_PICTURE);
    }

    public void onActivityResult(int requestCode, int resultCode,
Intent data) {
        super.onActivityResult(requestCode, resultCode, data);

        if (requestCode == SELECT_PICTURE && resultCode == RESULT_OK) {

            fileUri = data.getData();
            new DownloadImage(this, new DownloadImage.OnImageReady() {
                @Override
                public void file(File image) {


                }

                @Override
                public void base64(String imageBase64) {


                }

                @Override
                public void bytes(byte[] imageBytes) {
                    b  = BitmapFactory.decodeByteArray(imageBytes, 0,
imageBytes.length);

                    img.setImageBitmap(b);

                }
            }).pickImgFromGallery(fileUri);



        }

    }

    private void saveData(final Bitmap bitmap) {

        id  =
```

```
SessionManagerUser.getInstance(CreateNewDonationActivity.this).getId().
getId();
        final String myTitle=title.getText().toString().trim();
        final String myDetails=details.getText().toString().trim();
        final String myLocation=location.getText().toString().trim();


        if(TextUtils.isEmpty(myTitle)){
            title.setError("Enter you Data please");
            title.requestFocus();
            return;
        }
        if(TextUtils.isEmpty(myDetails)){
            details.setError("Enter you Data please");
           details.requestFocus();
            return;
        }
        if(TextUtils.isEmpty(myLocation)){
           location.setError("Enter you Data please");
            location.requestFocus();
            return;
        }
        VolleyMultipartRequest multipartRequest = new
VolleyMultipartRequest(Request.Method.POST,
                URLs.URL_ADD_DONATION,
                new Response.Listener<NetworkResponse>() {
                    @Override
                    public void onResponse(NetworkResponse response) {
                        String resultResponse = new
String(response.data);
                        try {

Log.d(CreateNewDonationActivity.class.getName(),"response"+response);
                            JSONObject result = new
JSONObject(resultResponse);
                            JSONObject object =
result.getJSONObject("data");


                            Toast.makeText(getApplicationContext(),
result.getString("message"), Toast.LENGTH_SHORT).show();


                            Intent intent = new
Intent(CreateNewDonationActivity.this,DonorMainActivity.class);
                            startActivity(intent);

                        } catch (JSONException e) {
                            e.printStackTrace();
                        }
                    }
                }, new Response.ErrorListener() {
            @Override
            public void onErrorResponse(VolleyError error) {

Toast.makeText(getApplicationContext(),error.toString(),
Toast.LENGTH_LONG).show();
```

```
                NetworkResponse networkResponse =
error.networkResponse;
                String errorMessage = "Unknown error";
                if (networkResponse == null) {
                    if (error.getClass().equals(TimeoutError.class)) {
                        errorMessage = "Request timeout";
                    } else if
(error.getClass().equals(NoConnectionError.class)) {
                        errorMessage = "Failed to connect server";
                    }
                } else {
                    String result = new String(networkResponse.data);
                    try {
                       // JSONObject response = new JSONObject(result);
                        // String status =
response.getString("status");
                        //   String message =
response.getString("message");

                        Log.e("Error Status", "status");
                        Log.e("Error Message", "message");

                        if (networkResponse.statusCode == 404) {
                            errorMessage = "Resource not found";
                        } else if (networkResponse.statusCode == 401) {
                            errorMessage = " Please login again";
                        } else if (networkResponse.statusCode == 400) {
                            errorMessage =  " Check your inputs";
                        } else if (networkResponse.statusCode == 500) {
                            errorMessage =  " Something is getting
wrong";
                        }
                    }// catch (JSONException e) {
                    catch (Exception e) {
                        throw new RuntimeException(e);
                    }
                    //e.printStackTrace();
                   // }
                }
                Log.i("Error", errorMessage);
                error.printStackTrace();
            }
        }){

            @Override
            protected Map<String, String> getParams() {
                Map<String, String> params = new HashMap<>();
                params.put("Content-Type","application/json");
                params.put("Accept","application/json");
params.put("Accept","application/json");
                params.put("donor_id",String.valueOf(id));
                params.put("title",myTitle);
                params.put("details",myDetails);
                params.put("location",myLocation);
                return params;
            }
```

```
            @Override
            protected Map<String, DataPart> getByteData() {
                Map<String, DataPart> params = new HashMap<>();
                // file name could found file base or direct access
from real path
                // for now just get bitmap data from ImageView
                long imagename=System.currentTimeMillis();
                params.put("image", new DataPart(imagename + ".png",
getFileDataFromDrawable(bitmap)));


                return params;
            }
        };


VolleySingleton.getInstance(CreateNewDonationActivity.this).addToReques
tQueue(multipartRequest);
    }
    public  byte[] getFileDataFromDrawable(Bitmap bitmap){
        ByteArrayOutputStream byteArrayOutputStream= new
ByteArrayOutputStream();
        bitmap.compress(Bitmap.CompressFormat.JPEG, 60,
byteArrayOutputStream);
        return byteArrayOutputStream.toByteArray();
    }
}
```

Chapter 6 System Testing

Ensuring that the application works correctly and provides services to all its users as required of it is one of the most important things that must be done. Therefore, in this section, the results of the tests that were conducted on the application will be presented.

6.1 Software Testing

Software testing means checking a program to see whether it works the way it is supposed to or not and ensuring that it meets the assumed requirements and is free of errors or problems and that it works well, is safe and reliable.

6.1.1 Software Validation

Software validation is a method of comparing a software product, to ensure that the software meets previously defined and described employer requirements as well as personal/customer requirements and expectations. It is specially conducted for the purpose of checking whether the advanced software is designed in line with the pre-defined Software Application Requirements Specifications (SRS) and whether it meets the real desires of the customers or not. Validation is usually done when software development is completed.

6.1.2 Software Verification

Software verification is one of the most important things that programmers must do in order to verify whether the program achieves the required goals and is free of errors, as all pre-defined software specifications are checked to ensure that the product meets them. It includes several activities such as checking system requirements, business requirements, code guidelines, and design review.

6.2 Usability Testing

Usability testing in software testing is a type of testing, which is conducted from the point of view of the end user to determine whether the system is easily usable and the practice of testing the usability of a design on a group of representative users. In usability testing, conducting it too late in the design process is a very common mistake.

Testators Tasks:

Table 6.1: Testators Tasks

Task-Number	Task Name	Excellent	Accepted	Unaccepted	Real time
1	Register	< 3 sec	4 – 5 sec	>5 sec	2 sec
2	Manage Profile	< 3 sec	4 – 5 sec	>5 sec	2 sec
3	Create a new will	< 3 sec	4 – 5 sec	>5 sec	2 sec
4	Enter a list of assets	< 3 sec	4 – 5 sec	>5 sec	2 sec
5	Appoint executors or representatives	< 3 sec	4 – 5 sec	>5 sec	2 sec
6	Request support	< 3 sec	4 – 5 sec	>5 sec	2 sec

Executors (Responsible for Applying the Will) Tasks:

Table 6.2: Executors (Responsible for Applying the Will) Tasks

Task-Number	Task Name	Excellent	Accepted	Unaccepted	Real time
1	Login	< 3 sec	4 – 5 sec	>5 sec	3 sec
2	Access the will documents and related information for the testator they are representing.	< 3 sec	4 – 5 sec	>5 sec	2 sec

Donors Tasks:

Table 6.3: Donors Tasks

Task-Number	Task Name	Excellent	Accepted	Unaccepted	Real time
1	Register	< 3 sec	4 – 5 sec	>5 sec	2 sec
2	Manage Profile	< 3 sec	4 – 5 sec	>5 sec	2 sec
3	Create a new will	< 3 sec	4 – 5 sec	>5 sec	2 sec
4	Enter a list items they want to donate	< 3 sec	4 – 5 sec	>5 sec	2 sec
5	schedule or request pickups or drop-offs for their donated items.	< 3 sec	4 – 5 sec	>5 sec	2 sec
6	communicate with recipient.	< 3 sec	4 – 5 sec	>5 sec	2 sec

Customer Support Tasks:

Table 6.4: Customer Support Tasks

Task-Number	Task Name	Excellent	Accepted	Unaccepted	Real time
1	Login	< 3 sec	4 – 5 sec	>5 sec	2 sec
2	View the sent support requests	< 3 sec	4 – 5 sec	>5 sec	2 sec
3	Respond to the support requests	< 3 sec	4 – 5 sec	>5 sec	2 sec

Unregistered user Tasks:

Table 6.5: Unregistered user Tasks

Task-Number	Task Name	Excellent	Accepted	Unaccepted	Real time
1	view donated items	< 3 sec	4 – 5 sec	>5 sec	2 sec
2	contact donors to get the donated items	< 3 sec	4 – 5 sec	>5 sec	2 sec

6.3 Test Cases

- **Test Case (1)**

Task Name: Register

Description: Testators can create an account in the application by entering their basic data.

Function:

1. Open the application
2. Go to the account creation page
3. Enter all fields
4. Press the Register button

Table 6.6: Test Case (1)

Testing case	Expected Output	Pass / Faild
Enter a password of less than 8 characters	An error message will appear telling the user that the number of characters in the password must not be less than 8 characters	Pass
Entering an incorrect email	A message will appear informing the user that the email he entered is incorrect	Pass

The user left one of the account creation form fields empty without a value	The system will not allow the account to be created and will mark the user with empty fields	Pass

- **Test Case (2)**

Task Name: login

Description: Both the donor and the will maker can log in to the application.

Function:

1. Open the application
2. Go to the login page
3. Enter all fields
4. Press the login button

Table 6.7: Test Case (2)

Testing case	Expected Output	Pass / Faild
Entering an incorrect email	A message will appear informing the user that the email he entered is incorrect	Pass
The user entered data while he did not have an account in the application.	The system will tell the user that his data is incorrect and that he does not have an account in the application, and a message will appear	Pass

	telling him that he must create an account first.	
The user left one of the account creation form fields empty without a value	The system will not allow the account to be created and will mark the user with empty fields	Pass

Conclusion

In this project, the theoretical analysis and practical implementation of the application concept was presented, which enables users to save their wills electronically, along with many other services, such as adding property and debts, publishing donations, and delivering them to people in need. Using this program, users can secure their will and ensure that it reaches the people they want. This project presented the difficulties that members of Saudi society face in keeping their will, as well as their desire to help people and deliver donations to them, as they were all solved in a safe and effective manner through this application.

Future Work

There are many future works to improve the functionality of the application such as:

- Linking the application to a website.
- Add the application evaluation feature.
- Linking with government databases
- Improving the application interfaces and making them more attractive.

Chapter 7 References

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